October 2023

1. Read preventions.

Q. Preventions :

1. Lithium in Bipolar to prevent suicide - **Tertiary Prevention**
2. At risk mental state boy – **Indicated (risk mental state)**
3. Suicide helpline – **Universal primary**
4. Antibullying school programme – **Universal**

[SPMM] Prevention in mental health

Choose the most appropriate term(s) from the options for each prevention scenario given below. Each option can be used more than once or not at all.

1. For individuals identified as having an "at-risk mental state" for psychosis, a prevention strategy is implemented offering Cognitive Behavioural Therapy (CBT) (choose TWO)
2. A school program aimed at providing educational and support services to low-income families to reduce the risk factors associated with mental health problems in children (choose TWO)
3. A one-day school-wide retreat program is designed to prevent bullying among students (choose TWO)

- Indicated prevention
- Primary prevention
- Secondary prevention
- Selective prevention
- Tertiary prevention
- Universal prevention
- Promotion

The first attempt to classify prevention began in the 1960s with Caplan's model.

Prevention level	Description	Examples
Primary Prevention	Aim is to intervene before a disease or problem begins. It aims to reduce the incidence of an illness in the community, and is directed at individuals who are at risk for developing a particular disorder. **It can be subdivided into universal, selective, and indicated prevention**	Cognitive interventions provided to adolescents with cognitive deficits years before the first psychotic episode to prevent the later phases of schizophrenia Prophylactic use of antipsychotics for sub-syndromal schizophrenia CBT after a traumatic event to reduce the incidence of PTSD in those who are at greatest risk **Measures to reduce the incidence of suicide**
Secondary Prevention	Secondary prevention aims to detect and treat disease that has not yet become symptomatic and therefore to lower the rate of established cases of the disorder. This is achieved through early detection and treatment	Screening procedures
Tertiary Prevention	Tertiary prevention involves the care of established disease, with attempts made to restore to highest function, minimize the negative effects of disease, and prevent disease-related complications	**The use of lithium to prevent episodes of mania in people with bipolar disorder**

In 1992 a new system was devised (by the Institute of Medicine) that held a more pure view of prevention. Prevention was thus redefined as only those interventions used before the initial onset of a disorder. Again there are three main levels.

Prevention level	Description	Examples
Universal Prevention	Targeted to the general public or a whole population that has not been identified on the basis of individual risk. The intervention is desirable for everyone in that group. Universal interventions have advantages when their costs per individual are low, the intervention is effective and acceptable to the population, and there is a low risk from the intervention	Healthy eating and exercise School-based programs offered to all children to teach social and emotional skills or to avoid substance abuse Programs offered to all parents of sixth graders to provide them with skills to communicate to their children about resisting substance use
Selective Prevention	Targeted to individuals or a population subgroup whose risk of developing mental disorders is significantly higher than average. The risk may be imminent or it may be a lifetime risk. Risk groups may be identified on the basis of biological, psychological, or social risk factors that are known to be associated with the onset of a mental, emotional, or behavioural disorder. Selective interventions are most appropriate if their cost is moderate and if the risk of negative effects is minimal or non-existent	Providing support for socially isolated elderly people Programs offered to children exposed to risk factors, such as parental divorce, parental mental illness, death of a close relative, or abuse, to reduce risk for adverse mental, emotional, and behavioural outcomes
Indicated Prevention	Targeted to high-risk individuals who are identified as having minimal but detectable signs or symptoms foreshadowing mental, emotional, or behavioural disorder, or biological markers indicating predisposition for such a disorder, but who do not meet diagnostic levels at the current time. Indicated interventions might be reasonable even if intervention costs are high and even if the intervention entails some risk	**Low-dose atypical antipsychotics and CBT for patients with prodromal symptoms of schizophrenia to delay or prevent disease onset** Interventions for children with early problems of aggression or elevated symptoms of depression or anxiety

Prevention Strategy	Description	Example
Universal Preventative Intervention	Designed to prevent problems in entire populations, regardless of individual risk factors	Public health campaigns, policies promoting healthy lifestyles
Selective Preventative Intervention	Targets specific subgroups of the population at increased risk of developing a problem	School-based prevention programs for at-risk youth targeted screening programs for certain health conditions
Indicated Preventative Intervention	Designed for individuals showing signs of a problem or at very high risk of developing a problem	Early intervention programs for children with developmental delays, targeted counselling for individuals with substance abuse problems

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Types of prevention I
Identify the mode of prevention for each of the situations given below;

Woman with history of depression stopped venlafaxine after her dad had a heart attack

ECG monitoring of patients on clozapine.

Use of rivastigmine in dementia patients.

Choose...
Secondary prevention
Tertiary prevention
Selective Primary prevention
Indicated primary prevention.
Quaternary prevention
Not a preventative measure

Your answer is incorrect.

Explanation: 1) Indicated Primary Prevention. Given her familial risk of a cardiovascular event is increased, but has not yet happened to her, this is a form of Primary Intervention, particularly indicated for her. 2) Strictly speaking, ECG monitoring is not a preventative measure 3) Tertiary Prevention. This is indicated in cases of chronic illness, with the aim of slowing down progression.

The correct answer is: Woman with history of depression stopped venlafaxine after her dad had a heart attack → Indicated primary prevention., ECG monitoring of patients on clozapine. → Not a preventative measure, Use of rivastigmine in dementia patients. → Tertiary prevention

Types of prevention II
Identify the mode of prevention used in each of the following situations;

A 22-year-old female recently diagnosed with schizophrenia is started on an antipsychotic medication

The provision of substance abuse education as part of the school curriculum for all school children in Englar

A 45-year-old schizophrenia patient is transferred from an acute ward to a rehabilitation unit following treatment of a psychotic episode.

Choose...
Primary prevention
Universal prevention
Tertiary prevention
Selective prevention
Secondary prevention

Your answer is incorrect.

Explanations: 1) Universal prevention strategies are designed to reach the entire population, without regard to individual risk factors and are intended to reach a very large audience. The program is provided to everyone in the population, such as a school or community. 2) Tertiary prevention focuses on reducing or minimizing the consequences of a disease once it has developed. The goal of tertiary prevention is to eliminate, or at least delay, the onset of complications and disability due to the disease. Most medical interventions fall into this category. 3) Secondary prevention attempts to identify a disease at its earliest stage so that prompt and appropriate management can be initiated. Successful secondary prevention reduces the impact of the disease

The correct answer is: A 22-year-old female recently diagnosed with schizophrenia is started on an antipsychotic medication → Secondary prevention, The provision of substance abuse education as part of the school curriculum for all school children in England → Universal prevention, A 45-year-old schizophrenia patient is transferred from an acute ward to a rehabilitation unit following treatment of a psychotic episode. → Tertiary prevention

2. Personality disorders (Anankastic, Schizoid, bipolar Ratio 1:1)
   Prevalence personality disorder in general population (5 -13%) OCPD 2 – 8%

Disorder	**Male:Female Ratio**
OCPD	1:1
Avoidant	1:1.5
Dependant	1:1.5
Histrionic	1:1.8
Borderline	1:3
Antisocial	5:1
Paranoid	1.3:1

Disorders with higher female preponderance		
Disease	**Ratio**	**Higher in**
Social phobia (May be equal in those seeking treatment)	1.2:1	Females
Bipolar disorder type 2	1.3:1	Females
Avoidant personality	1.5:1	Females
Dependent personality	1.5:1	Females
OCD (May be equal in those seeking treatment)	1.5:1	Females
Histrionic personality	1.8:1	Females
Alzheimer's disease (prevalence rates 2:1. Incidence may be equal. More female representation may be due to longevity of lives in women).	2:1	Females
Dysthymia	2:1	Females
Generalised anxiety disorder	2:1	Females
Major depression (this is post-puberty prevalence rates; before puberty, boys marginally > girls)	2:1	Females
Multiple Sclerosis	2:1	Females
PTSD	2:1	Females
Somatization disorder	2:1	Females
Conversion disorder	2-10:1	Females
Panic disorder	2-2.5:1	Females
Agoraphobia	3:1	Females
Borderline personality	3:1	Females
Specific phobias	3:1	Females
Nightmares	2-4:1	Females
Dissociative disorders	4:1	Females
Bulimia nervosa (refer Kaplan & Sadock CTP – quotes new estimates 3:1)	10:1	Females
Anorexia nervosa	10:1	Females

Disorders with higher male preponderance		
Disease	**Ratio**	**Higher in**
Lewy body dementia	1.2:1	Males
Paranoid personality disorder	1.3:1	Males
Schizophrenia (Saha et al PLoSMed 2004) incidence rates only. Prevalence is equal.	1.4:1	Males
Parkinson's disease	1.5:1	Males
Conduct Disorder		Males
ADHD	3:1	Males
Alcohol dependence	5:1	Males
Korsakoff's psychosis	2:1	Males
Autism	3:1	Males
Asperger's syndrome **Wing 1981 = 10:1**	4:1	Males
Antisocial personality disorder	5:1	Males
Tourette's syndrome	4:1	Males

Disorders with equal sex distribution		
Disease	**Ratio**	**Higher in**
Anankastic (OCP) personality	1:1	Neither
Frontal Lobe Dementia	1:1	Neither
Huntington's disease (AD inheritance)	1:1	Neither
Schizoid personality disorder	1:1	Neither
Wilson's disease (puzzling 3:1 female preponderance seen for acute liver failure)	1:1	Neither
Bipolar disorder type 1	1:1	Neither

3. Read Tay Sachs: Rare **autosomal recessive disorder**, **more common among Ashkenazi Jews**, **Deficiency of Beta N acetylhexosamindase A**

Inheritance pattern	Examples
Autosomal Dominant	Neurofibromatosis type 1 and 2, tuberous sclerosis, achondroplasia, Huntington disease, Noonan's syndrome
Autosomal Recessive	Phenylketonuria, homocystinuria, Hurler's syndrome, galactosaemia, **Tay-Sach's** disease, Friedrich's ataxia, Wilson's disease, cystic fibrosis
X-Linked Dominant	Vitamin D resistant rickets, Rett syndrome
X-Linked Recessive	Cerebellar ataxia, Hunter's syndrome, Lesch-Nyhan
Mitochondrial	Leber's hereditary optic neuropathy, Kearns-Sayre syndrome

4. **Phenylketonuria PKU**:

- **Autosomal recessive disorder**,
- Caused by deficit of **phenylalanine hydroxylase (PAH)**
- **Characterised by:**
  - Microcephaly.
  - Language delay.
  - Hypopigmentation of skin.
  - Hyperactivity.
  - Severe ld.
  - Self-injury.
  - Musty or mouse-like odour.
  - Management diet: low protein diet, limit food with phenylalanine, sapro-protein.

[SPMM] A four-year-old boy who was born in India presents with epilepsy, severe mental retardation and behavioural problems. On examination, you note the lack of pigment in skin and hair. The diagnosis is

Select one:

A. Lesch Nyhan syndrome
B. Phenylketonuria
C. Velocardiofacial syndrome
D. Williams syndrome
E. Taysachs disease

Explanation: Phenylketonuria (PKU) is an autosomal recessive metabolic genetic disorder. A defect in Phenylalanine Hydroxylase (PAH) or cofactor (biopterin) with accumulation of phenylalanine. Symptoms absent neonatally, a later development of seizures (25% generalized). Clinical features include fair skin, blue eyes, blond hair, and recurrent rashes. Untreated infants have an unusual mouse-like body odour. Microcephaly is seen in 50% of cases. Without treatment mild to profound mental retardation, language delay, destructiveness, self-injury, hyperactivity could result. A low-phenylalanine diet must be usually continued throughout the childhood.

5. Korsakoff which test?? – There are no specific test except for clinical diagnosis, triad of symptoms – ophthalmoplegia, ataxia and confusion.

Q. from another recall version, Korsakoff - what test is useful : *(Depend on other options)*

A. Address and recall,
B. Digit span,
C. Few other options

[SPMM] Robert has a history of chronic alcoholism and now diagnosed with Korsakoff's psychosis. Which of the following tests will **NOT** be impaired?

Select one:

A. Abbreviated mental test
B. Minnesota Multiphasic personality inventory
C. Digit span
D. MMSE
E. WAIS

Patients with Korsakoff psychosis **perform well on Digit span subtest** of WAIS. But due to episodic memory loss, they **lose scores on recall test** in MMSE and AMT. In addition, a degree of frontal dysfunction results in a reduction in overall WAIS scores, which may recover over a period of time if the patient remains abstinent from alcohol. Interestingly, amnesic patients show elevated scores in several domains of MMPI, highlighting the frontal damage effects of alcohol abuse. In addition, MMPI also has a substance abuse scale in it.

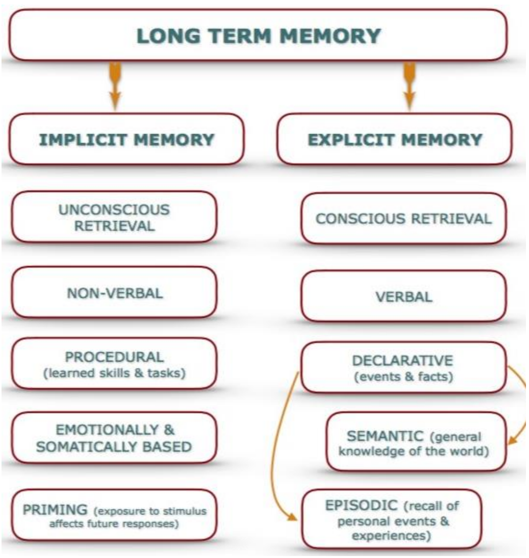
6. Is it **explicit memory** affected in ECT? Implicit, Episodic and semantic affects both Antero and retrograde amnesia but retrograde more

[SPMM] What type of memory is most affected by ECT?

Select one:

A. Implicit memory
B. Retrograde memory
C. Semantic memory
D. Episodic Memory
E. Anterograde memory

Retrograde memory **(Explicit memory)** i.e. the recall of events that occurred before the treatment, is the most commonly affected aspect of memory following Electro-Convulsive Therapy (ECT). It is an early side effect, occurs in **64%** of patients receiving ECT and usually resolves completely.



***N.B: The sole absolute contraindication for ECT is a known phaeochromocytoma**. Increased intracranial pressure, current intracranial masses, recent stroke, cardiac conduction abnormalities, high risk pregnancy, aortic and cerebral aneurysms, asthma/COPD treated with theophylline are relative contraindications.* [Fresh Questions(set 5) Q.71]

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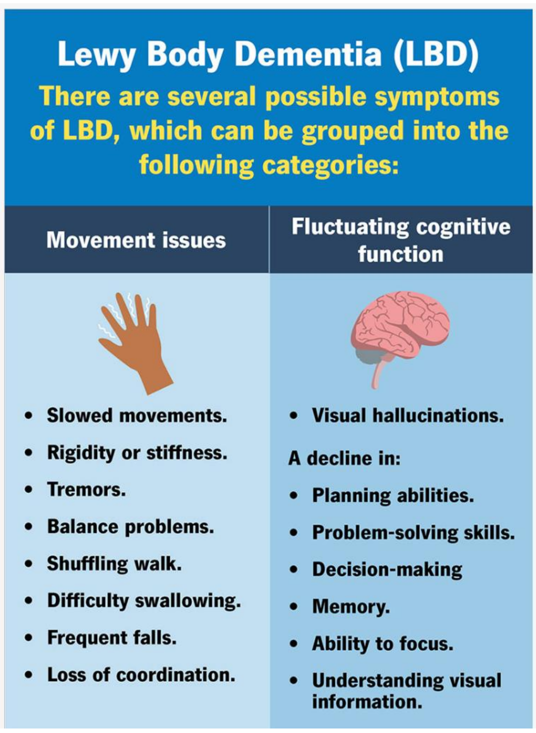
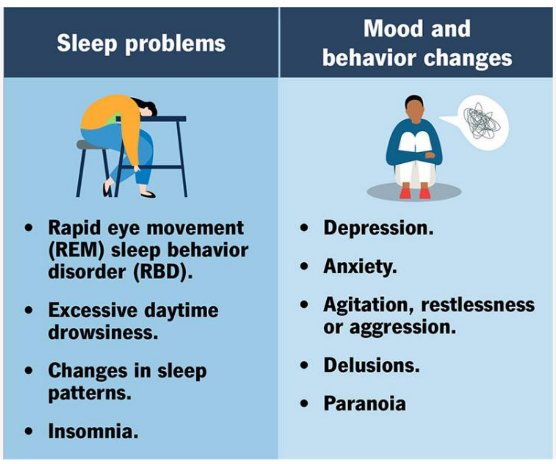
7. Mild depression - **2 weeks of watchful waiting**

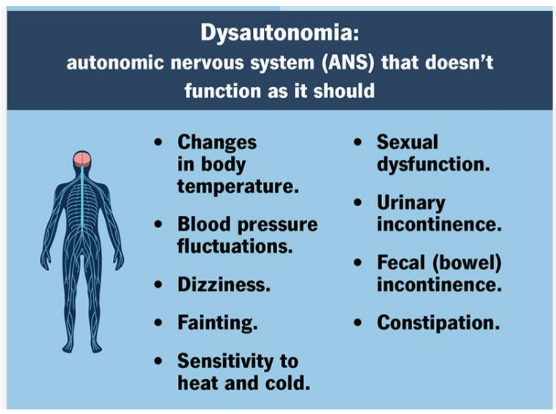
Severity	**PHQ-9**	**Comment**	**First-line options**
Less severe (mild and subthreshold)	< 16	Subthreshold and mild depression	Guided self-help Group or individual CBT Group or individual behavioural activation Group exercise Group mindfulness and meditation Interpersonal psychotherapy (IPT) **SSRIs** Counselling Short-term psychodynamic psychotherapy (STPP)
More severe (moderate and severe)	> 16	Moderate and severe depression	Combined CBT plus antidepressant Individual CBT Individual behavioural activation Antidepressants Individual problem solving Counselling Short-term psychodynamic psychotherapy (STPP) Interpersonal psychotherapy (IPT) Guided self-help Group exercise

8. Fluctuating memory loss – **DLB**

Lewy Body Dementia (LBD): accounts for 15-20% of cases of dementia in hospital and community based samples.







9. Paroxetine - **Cardiac malformation (ASD/VSD)**

    [SPMM] Q. A 27-year-old woman, who is 5 months pregnant now, is now depressed. She has a past and a family history of depression. She had a good response to paroxetine and insists to continue paroxetine during this episode. What adverse effect will you warn her about?

    Explanation: Paroxetine, particularly high dose first trimester exposure, is clearly linked to **cardiac malformation - VSD and ASD.** Third trimester use can give rise to neonatal complication due to abrupt withdrawals. Exposure to SSRI, when taken in late pregnancy, may increase risk of persistent pulmonary hypertension of the new born, although the absolute risk is small.

10. Cannabis q - healthy entrant bias

    Healthy entrant bias ??? "Healthy entrant bias" refers to the tendency for participants in studies or programs to be healthier than the general population. This bias can impact the validity of findings, as healthier individuals may respond differently to interventions than those who are less healthy.

11. Use of threats and violence to punish, harm or intimidate - **coercive control**.

12. OCD gender ratio **1:1**

13. Immature defence mechanism - **Identification** (projective identification but was not in the options)

Immature defences (Level 2). These include:-

Defence	Notes
Projection	Attributing uncomfortable thoughts or feelings to others
Acting out	Acting on thoughts or emotions forbidden by the superego
Projective identification	the object of projection invokes in that person precisely the thoughts, feelings or behaviours projected

Mature defences (Level 4). These are considered the most advanced form of defence mechanisms and include:-

Defense	Notes
Altruism	Lessening negative feelings by providing constructive service/generosity to others
Sublimation	Redirecting negative thoughts or feelings into a more positive form
Suppression	Process of consciously avoiding thinking about something for example by distracting oneself
Humour	Using ideas and feelings that are pleasurable to others to alleviate a challenging situation
Identification	**The unconscious modelling of one's self upon another person's character and behaviour**

Psychotic defences (Level 1). These are said to be pathological as they distort experiences in such a way that they eliminate the need to deal with reality. They include:-

Defense	Notes
Denial	Refusal to acce pt reality
Distortion	A gross reshaping of reality to meet internal needs
Splitting	Intolerance of ambiguity leading to self/others being perceived as being wholly good or bad

Neurotic defenses (Level 3). These tend to have short term advantages but lead to problems when used in the longer term. They include:-

Defense	Notes
Displacement	Redirection of impulses onto a different target to the one who caused the emotion
Reaction formation	Acting in the opposite way to the thought or feeling
Repression	Process of keeping unwanted emotions or thoughts outside conscious awareness
Intellectualization	Focusing on details in an effort to avoid painful thoughts or emotions
Dissociation	Temporary drastic modification of one's personal identity or character to avoid emotional distress
Isolation	The disconnection of an event from the emotion attached to it
Regression	Reverting back to an earlier stage of development when faced with an unpleasant thought or emotion
Rationalization	The creation of false but credible justifications
Undoing	An attempt to take back an unpleasant thought or emotion

Mature (CA^3S^2H)
Compensation
Altruism
Asceticism
Anticipation
Suppression
Sublimation
Humour

Immature ($D^2I^2P^3$ SAR)
Denial
Denigration
Introjection
Idealization
Passive aggression
Projective identification
Splitting
Acting out
Regression

SIP DOG -

14. What was the answer for incremental QALY? – **8** [It will be a calculation question] (economic q)

15. CAT initiation phase – **exploration**

- **Initial phase:** an **exploration** of traps, dilemmas and snags. Therapist writes formulation letter
- **Middle phase:** working through problems with the use of diagrams exploring 'target problem procedures.'
- **Ending phase:** both patient and therapist write goodbye letters

16. what kind of study was it? CEA? or **CUA**? The QALY questions. 8 and cost utility study? Something like equal outcome at Lower cost??

[SPMM] A patient with schizophrenia, with a future life expectancy of 40 years, is residing in a residential accommodation for the last 10 years. It is estimated that this residential placement offers a QALY of 0.5 and costs £900 a week, while an alternative option across the road can improve the QALY to 0.7, but costs £1100 a week. What type of economic analysis is being conducted?

Select one:

A. Cost consequences
B. Cost effectiveness
C. Cost minimisation
D. Cost utility
E. Cost benefit

Explanation: QALYs provide a common unit to assess the benefit of various interventions in terms of health- related quality of life as well as survival. In economic analysis, the use of QALYs along with the costs of interventions is called cost–utility analysis.

[SPMM]A patient with schizophrenia, with a future life expectancy of 40 years, is residing in a residential accommodation for the last 10 years. It is estimated that this residential placement offers a QALY of 0.5, and costs £900 a week. An alternative residential care home across the road can improve the QALY to 0.7, but costs £1100 a week. How many QALYs could the patient expect, if she were to move to the new residential facility?

Select one:

A. 40 years
B. 28 years
C. 32 years
D. 20 years
E. 36 years

Explanation: This question asks you to calculate the total QALYs over the expected lifetime at the new facility. At the new home, if 40 years are spent in health state 0.7, the number of QALYs generated = 40*0.7 = 28 years.

17. **Smith Magenis scenario**

[SPMM] Alfie Mordant is a 6 year old boy with moderate intellectual disability who tends to be sleepy during the day but with disturbed sleep at night. His parents are very worried about his behaviour as he can cause injury to himself. At other times he hugs himself repeatedly.

Explanation: This pattern of self-hugging, self-injurious behaviour and sleep disturbance in the presence of moderate intellectual disability is most associated with Smith Magenis Syndrome. **The presence of self-hugging is a particular feature of this disorder. It is associated with a complete or partial deletion of 17p11.2**

Smith Magenis syndrome	Hyperactive, impulsive, aggressive, stereotypic movements (self-hugging) and self injury such as head banging, nail pulling, putting objects into bodily orifices etc.

Characteristic features:

- Flattened mid face
- Abnormally shaped upper lip
- Short hands and feet

- Single transverse crease
- Abnormally placed ears
- Protruding tongue
- High arched palate
- Small toes
- Hoarse, deep voice

18. **Lesh Nyan scenario**

[SPMM] A 2-year-old boy with microcephaly and intellectual disability has developed severe self-harm behaviours. He also displays spasticity and choreiform movements. What is the most likely diagnosis?

Select one:

A. Lesch-Nyhan syndrome
B. Hunter syndrome
C. Hurler syndrome
D. Fragile X syndrome
E. Rett syndrome

Explanation: Lesch-Nyhan syndrome is an extremely rare **X-linked recessive** condition due to a mutation in the **HPRT gene**, leading to hyperuricaemia. There is development of spasticity, choreiform movements, and transient hemiparesis. Microcephaly is common and intellectual disability is usually severe. Behavioural problems include **self-mutilation (biting of lips, inside of mouth, fingers). It may be associated with verbal and physical aggression.**

19. How about those paraesthesia Qs

20. **Hyperesthesia**

Increased sensibility to stimuli of sense. [ICD11]

21. **Allodynia**

Pain due to a normally non-painful stimulus. [ICD11]

22. **Hypoesthesia**

Decreased sensibility to stimuli of sense.

23. Regarding Lithium in pregnancy question was there an option to stop on day of delivery and check 24 hrs after? **Instead, the option was decrease the dose of lithium to 400 and check one week later.**

[SPMM] A pregnant woman with bipolar I disorder is scheduled for induction of labour next week. Her lithium dose has been increased in recentweeks, and she has a current level of 0.75 mmol/L. She is not planning to breastfeed. What is the most advisable course of management?

Select one:

A. Suspend lithium 24-48 hours before induction and re-check the level 12 hours after the last dose
B. Reduce the lithium dose to 400 mg and re-check the level 1 week post-delivery
C. Maintain current lithium dosing and re-check the level 4 weeks post-delivery
D. Check the lithium level daily until her induction, then discontinue lithium until 4 weeks post-delivery

E. As she is not planning to breastfeed, no changes in lithium dosing or further bloodwork are advised

In the third trimester, the use of lithium may be problematic because of changing pharmacokinetics: an increasing dose of lithium is required to maintain the lithium level during pregnancy as total body water increases, but requirements return abruptly to pre-pregnancy levels immediately after delivery. To prevent maternal and neonatal intoxication, lithium should be suspended for 24-48 hours before a planned Caesarean section or induction, and at the onset of labour, and the lithium level should be measured 12 hours after the last dose. If levels are not above the therapeutic range, lithium should be restarted on day 1 postnatal and the level checked again after 1 week.

[SPMM] Mrs Yanica Presley is a 32-year-old woman with a previous diagnosis of bipolar affective disorder and has had several previous episodes of mania requiring admission. She has responded well to Lithium and has had no relapses since it was started 3 years ago. She is currently 36 weeks pregnant and has been taking Lithium 600mg throughout her pregnancy. She is currently well in her mood and is not planning on breastfeeding. Which of the following would be the most appropriate action? [fresh Q. set 5]

Select one:

A. Stop lithium immediately
B. Do nothing and review after 1 week after delivery
C. Increase dose to 800 mg
D. Reduce dose to 400 mg, then repeat blood levels after 1 week
E. Continue lithium and check levels weekly until delivery and for 2 weeks after

Lithium can be continued throughout pregnancy provided that adequate monitoring is provided. This is usually 4 weekly until 34 weeks and then weekly until 2 weeks post-partum, with doses being modified as appropriate. **An increase in dose is often required earlier in the pregnancy with this reversing to an extent in late pregnancy**. Lithium can be omitted for 24-48 hrs prior to labour to reduce effects on the new born but this must be balanced against the risk of relapse in high risk women. Women at high risk of relapse in the post-partum must not be left without effective treatment.

Lithium treatment programme in pregnant mothers (Adapted from Kohen,2004)
• In women on maintenance treatment, serum lithium level should be monitored every 4 weeks through out the pregnancy • Lithium dosage should be adjusted to match the lower end of the therapeutic range • Lithium should not be discontinued abruptly; prior to delivery the dosage should be gradually tapered to 60-70% of the original maintenance level • Lower doses and frequent blood monitoring should be norm in pregnant women starting lithium in the first trimester of pregnancy • Lithium commenced in second and third trimester of pregnancy or perinatal period can help reduce the risk of puerperal psychosis • No decision is risk free • guidelines should vary with the severity of the illness • Should undergo level 2 ultrasound and echocardiography of the foetus at 6 & 18 week's gestation to screen Ebstein's anomaly.(Maudsley.2007) • Increased dose of lithium is required during 3rd trimester as total body water increases, but the requirement returns abruptly to pre-pregnancy level immediately after delivery.

24. What was the first line treatment for Oppositional defiant disorder? **Parenting skills training**

Management: NICE recommend the following in the treatment of antisocial behaviour, oppositional defiant disorder and conduct disorders in children and young people:- (Note there is some overlap)

Age	Intervention
3-11	**Group parent based training programs** (or parent AND child training programmes for children with complex needs). If both parents not able to participate then individual training should be offered
9-14	Child-focused programmes
11-17	Multimodal interventions with a family focus

[SPMM]NICE recommends that group based **parent training/education programmes should be the mainstay of treatment for children of 12 years and under with oppositional defiant disorder and conduct disorder.**

25. **Malingered PTSD -** Exact same reoccurring dream - theme remains same in PTSD but dream varies.

    Malingering of PTSD is the intentional production of false or grossly exaggerated physical and/or psychological symptoms associated with the diagnosis of PTSD in order to obtain external incentives (e.g., financial and/or personal gains).

26. Pathological lying: **"lie with the matrix of truth" some lie some truth**
    Pseudologia fantastica
    knowledge was in fictitious disorder question.

    [SPMM]Identify the true statement regarding pseudologia fantastica

    Select one:

    A. Absence of speech without impairment of consciousness
    B. An exaggerated sense of one's own importance
    C. Reality intermixed with fabricated material
    D. Episode of discrete amnesia related to alcohol intoxication
    E. A form of delusional belief shared by 2 close people

    Explanation: Pseudologia fantastica is one of the more extreme forms of pathological lying. This is a matrix of facts and fiction, mixed together in a way that makes the reality and fantasy almost indistinguishable. It is usually seen in forensic practice.

    [Fish psychology] **Pseudologia fantastica or fluent plausible lying** (**pathological lying**) is the term used, by convention, to describe the confabulation that occurs in those without organic brain pathology such as personality disorder of antisocial or hysterical type. Typically the subject describes various major events and traumas or makes grandiose claims and these often present at a time of personal crisis, such as facing legal proceedings. Although it seems that the person with pseudologia believes their own stories and there is a blurring of the boundary between fantasy and reality, when confronted with incontrovertible evidence these individuals will admit their lying. Minor varieties of this occur in those who falsify or exaggerate the past in order to impress others. aka mythomania

27. The young boy with mild depression: 2 week follow up, Start anti-Depressants. Or CBT This was **group CBT.** Or digital cbt? GP has no appointments (**start with group and then individual if needed**)

Mild depression:	**Self-help, exercise, CBT, active monitoring and follow up.**
Moderate depression:	**Antidepressant + CBT, add antipsychotics if with psychotic symptoms.**
Severe depression:	**As in Rx for moderate**

28. Which medication made the hallucinations worse in the patient with parkinsonism?? **Levodopa?**

Most patients treated with Levodopa or dopamine agonists (Eg **Apomorphine, Ropinirole** etc) develop neuropsychiatric side effects such as visual hallucinations (most common), psychosis, anxiety, euphoria, mania, impulsive behaviour and delirium.

Neuropsychiatric complications of Parkinson's disease
For each of the following Parkinsonian conditions below, match the most commonly associated neuropsychiatric complication.

An 84-year-old gentleman with poor education, with late onset Parkinson's disease exhibits severe extra pyramidal signs	Choose...
A 76-year-old gentleman with late-onset Parkinson's disease on high dose levodopa and selegeline still exhibits several extrapyramidal signs	Choose...
A 73-year-old gentleman with a previous history of bipolar affective disorder is on levodopa and pramiprexole	Choose...
A 63-year-old woman with prominent right-sided extrapyramidal signs, notable bradykinesia and gait disturbances. She is significantly impaired in activities of daily living.	Choose...

Your answer is incorrect.

Explanation:

The risk factors for **cognitive impairment** in patients with Parkinson's disease include older patients, late-onset parkinson's disease, poor educational achievements and severe extrapyramidal signs.

Psychosis in Parkinson's disease is mostly drug related. Dopaminergic drugs such as levodopa and Selegiline can cause psychotic symptoms such as paranoia and hallucinations-especially visual modality.

Pramiprexole can cause exacerbation of mood symptoms and could precipitate **hypomania**/mania. Denovo mania is rare in Parkinson's. It is mostly associated with levodopa and dopamine agonist treatment especially in pre-existing bipolar affective disorder/ family history of bipolar disorder. Levodopa can result in pathological gambling, hypersexuality and hallucinations in addition to hypomania.

The risk factors for developing **depression** in parkinson's disease include female sex, younger age of onset, prominent right-sided signs, bradykinesia and gait disturbance, rapid disease progression, poorer cognitive status and activities of daily living

The correct answer is: An 84-year-old gentleman with poor education, with late onset Parkinson's disease exhibits severe extra pyramidal signs → Cognitive impairment, A 76-year-old gentleman with late-onset Parkinson's disease on high dose levodopa and selegeline still exhibits several extrapyramidal signs → Psychosis, A 73-year-old gentleman with a previous history of bipolar affective disorder is on levodopa and pramiprexole → Hypomania, A 63-year-old woman with prominent right-sided extrapyramidal signs, notable bradykinesia and gait disturbances. She is significantly impaired in activities of daily living. → Depression

29. The 400 participants and trial of some medication anti-depressant and then placebo

[SPMM] **Evaluating a diagnostic test for clinical use**

Answer the following questions on the basis of the information provided in the précis. A new diagnostic test is being introduced to detect schizophrenia, and s study reporting data on comparison with gold standard DSM-5 based consensus criteria is published evaluating 400 outpatient subjects in total. The new scale indicated schizophrenia as a diagnosis for 40 out of the 400 patients while the gold standard procedure identified schizophrenia in only 24 out of those 40. 8 patients identified to have schizophrenia using the gold standard test were missed by the new test.

1. What is the positive predictive value of the new test?
2. What is the sensitivity of the new test?
3. What is the specificity of the new test?

Explanation: Let us first parse out the numbers provided to us.

- "evaluating 400 outpatient subjects in total" - This refers to the final TOTAL box
- ."The new scale indicated schizophrenia as a diagnosis for 40 out of the 400 patients" - This refers to the TOTAL test positives box
- "the gold standard procedure identified schizophrenia in only 24 out of those 40" - This refers to the TRUE POSITIVE box
- " 8 patients identified to have schizophrenia using the gold standard test were missed by the new test" - This refers to the FALSE NEGATIVE box

Using the information provided, please create the following table
(Note: italicized font in this table refers to numbers that we computed, and not provided in the question)

	DSM +VE Schizophrenia	DSM -VE NO Schizophrenia	TOTAL
Test POSITIVE	24	16	40
Test Negative	8	352	360
TOTAL	32	368	400

Now,
PPV = true positives/ total positives expressed as a percentage. PPV = 24/40 = 60%
Sensitivity = true positive/ total diseased -= 24/32 = 75%
Specificity = true negatives/ total non-diseased = 352/368 = 95.7%

Correct Answer is :
1. What is the positive predictive value of the new test? - 60%
2. What is the sensitivity of the new test? - 75%
3. What is the specificity of the new test? - 95.7%

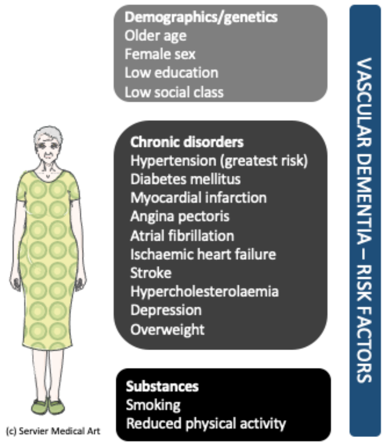
30. Was **vascular dementia** answer of any question? About Hyperlipidaemia, hypertension??

[SPMM] The greatest risk factor for vascular dementia is:

Select one:

A. Hyperlipidaemia
B. Ischaemic heart disease
C. Hypertension
D. Smoking
E. Alcohol consumption

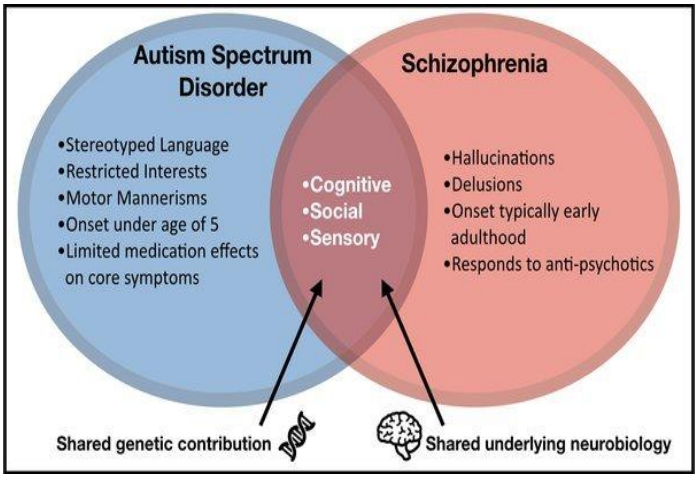
Explanation: The greatest risk factors for stroke and white matter hyperintensities and, consequently, for vascular dementia are high systolic and high diastolic blood pressure.



Risk Factors for vascular dementia

- Old age, Hypertension (50% of patients giving a positive history), Ischaemic heart disease
- Smoking, Alcohol consumption
- Hyperlipidemia, Atrial fibrillation, Family history, Valvular disease, Atrial myxoma, Carotid artery disease
- APOE4 allele
- Polycythemia, Sickle cell anaemia, Coagulopathies

31. Differentiate between schizophrenia and autism in child



32. There was this question with auditory hallucinations singing song ID Woman etc. The options were schizophrenia. epilepsy?

33. 2(two) hrs baby – crying , jittery
Few of the EMis were about maternal. Cocaine Maternal heroin. Maternal depression

Substance	Effect
Alcohol withdrawal:	Seen within 3-12 hours after delivery. Hyperactivity, crying, irritability, poor sucking, tremors, seizures, poor sleeping patterns, hyperphagia, and diaphoresis. Signs usually appear at birth and features of Foetal Alcohol Syndrome may be apparent later on further development.
Barbiturate withdrawal	Irritability, severe tremors, hyperacusis, excessive crying, vasomotor instability, diarrhea, restlessness, increased tone, hyperphagia, vomiting, and disturbed sleep.
Marijuana withdrawal	A mild opiate-like withdrawal syndrome with fine tremors, hyperacusis, and a prominent Moro reflex; mild and very rarely require treatment.
Nicotine withdrawal	Mild signs such as fine tremors and variations in tone; recent data indicates subtle neonatal behaviours, such as poor self-regulation and an increased need for handling.
Opiate withdrawal	Usually begins 24-48 hours after birth, depending on the time of last dose. However, signs may not appear in the infant until 3-4 days after birth. Hyperirritability, gastrointestinal dysfunction, respiratory distress, and vague autonomic symptoms (e.g., yawning, sneezing, mottling, fever). Tremors and jittery movements, high-pitched cries, increased muscle tone, and irritability are common. Normal reflexes may be exaggerated. Loose stools are common, leading to possible electrolyte imbalances and diaper dermatitis. Methadone withdrawal symptoms typically appear within 48-72 hours but may not start until the infant is aged 3 weeks. Milder with buprenorphine withdrawal.
Amphetamines and cocaine	Whether stimulants use causes, neonatal withdrawal remains unclear. Increasing evidence suggests that it is associated with an increased risk of prematurity and intrauterine growth restriction during pregnancy.
Antidepressants	Infants have been noted to show jitteriness, respiratory distress, and other neonatal complications. They are more commonly observed in newborns whose mothers were taking a short-acting SSRI such paroxetine and less with sertraline.

[SPMM]A new born baby is observed to be jittery, shaking, and **tachypnoeic**. Which of the following drugs taken by the mother is most likely responsible?
Select one:

A. Cocaine
B. Diazepam
C. Nicotine
D. Heroin
E. Alcohol

Explanation: Stimulant use (such as cocaine) prior to delivery is associated with a neonatal withdrawal syndrome including symptoms of agitation, vomiting, and **tachypnoea**.

[SPMM]**Neonatal drug effects**
Given the clinical picture below, identify the most appropriate diagnoses from list of options attached:

A. Nicotine Withdrawal
B. Floppy baby syndrome
C. Neuroleptic malignant syndrome
D. Alcohol withdrawal syndrome
E. Foetal alcohol Syndrome
F. Opiate withdrawal
G. SSRI discontinuation
H. Cannabis Withdrawal

1. Newborn baby with low birth weight, small head circumference and dysmorphic facial features
2. Newborn baby who is very irritable, high-pitched cry, cries a lot with shakiness, poor feeding, Sleep wake abnormalities and vomiting
3. Newborn shows mild tremors, intermittent irritability and **increased need for handling**. Neonatal reflexes are normal.
4. Newborn baby with low birth weight, breathing difficulties, **hypotonic** muscles and irritability.

Your answer is correct.

Explanation:

1) Foetal alcohol syndrome is the commonest non-genetic cause of learning disability. Intellectual impairment, dysmorphic facial features in the form of low birth weight, small head circumference and disruptive behaviour is noted.

2) Opiate withdrawal: It usually begins 24-48 hours after birth, depending on the time of last dose. The essential features are hyperirritability, gastrointestinal dysfunction, respiratory distress, and vague autonomic symptoms (e.g., yawning, sneezing, mottling, fever). Tremors and jittery movements, high-pitched cries, increased muscle tone, and irritability are common. Normal reflexes may be exaggerated. Loose stools are common, leading to possible electrolyte imbalances and diaper dermatitis.

3) Nicotine withdrawal: Mild signs such as fine tremors and variations in tone; recent data indicates subtle neonatal behaviours, such as poor self-regulation and an increased need for handling.

4) Floppy baby syndrome: Babies exposed to benzodiazepines during pregnancy are at risk for the following- breathing difficulties, low birth weight, floppy muscles, unstable body temperature, alterations in heart rate/function, irritability, convulsions, etc. Prenatal exposure to Benzodiazepine can cause toxicity and withdrawal effects at birth. Flumazenil, a Benzodiazepine receptor antagonist, can reverse many effects of BDZs and has been used in emergencies in some BDZ-exposed neonates, successfully reversing most of the above adverse effects. (Ref: Retrieved from http://emedicine.medscape.com/article/978492-clinical)

The correct answer is: Newborn baby with low birth weight, small head circumference and dysmorphic facial features → Fetal Alcohol Syndrome, Newborn baby who is very irritable, high-pitched cry, cries a lot with shakiness, poor feeding, Sleep-wake abnormalities and vomiting → Opiate withdrawal, Newborn shows mild tremors, intermittent irritability and increased need for handling. Neonatal reflexes are normal. → Nicotine withdrawal, Newborn baby with low birth weight, breathing difficulties, hypotonic muscles and irritability → Floppy baby syndrome

34. Did the question has epilepsy movements? something like lips smacking and abnormal shaking. But what about auditory hallucinations, do they also occur in **catamenial epilepsy**? Voices outside the head. Closest for me was catamenial epilepsy. Or does pms present with these features?

[SPMM] A 34-year-old woman being treated for seizures has irregular menstrual periods, at which time her seizures worsen. It is most likely that shehas which of the following?

Select one:

- **A. Catamenial epilepsy**
- B. Interictal anxiety
- C. Interictal psychosis
- D. Post-ictal depression
- E. Rolandic epilepsy

Catamenial (menstrual) epilepsy describes a worsening of seizures in relation to the menstrual cycle and may affect around 40% of women with epilepsy. Catamenial epilepsy has a significant negative impact on quality of life. There is uncertainty regarding which treatment works best and when in the menstrual cycle treatments should be taken.

[ChatGPT] Catamenial epilepsy is a type of epilepsy in which seizures are associated with the menstrual cycle. Seizure frequency tends to increase during specific phases of the menstrual cycle, such as ovulation or menstruation. Hormonal fluctuations, particularly changes in estrogen and progesterone levels, are believed to play a role in triggering these seizures. Management may involve adjusting antiepileptic medications or exploring hormonal therapies to better control seizures in relation to the menstrual cycle. .

[SPMM] Clinical criteria for PMDD. (Adapted from DSM5): In most menstrual cycles during the past year, at least 5 of the following 11 symptoms (including at least 1 of the first 4 listed) should present:

- **Depressed mood**
- **Marked anxiety**
- **Marked affective lability**
- **Marked anger or irritability (usually the most severe and start earlier)**
- Anhedonia
- Subjective sense of difficulty in concentrating
- Lethargy, easy fatigability, or marked lack of energy
- Marked change in appetite, overeating, or specific food cravings
- Hypersomnia or insomnia
- A subjective sense of being overwhelmed or out of control
- Physical symptoms, e.g., breast tenderness or swelling, headaches, joint or muscle pain, a sensation of bloating, or weight gain

35. Paroxetine causes? This question asked what the most common sign of paroxetine if taken during pregnancy.

A. PPHTN.
B. Irritability.
C. Ebstein's.

36. Conduct disorder is it parent skill training and management is same for ODD and CD **Parent skills training.**

Parent management training is the best established approach for managing children with conduct disorders and it is backed up numerous RCTs. Parents are taught to pay attention to desired behaviour. They are also taught effective techniques for handling undesirable behaviour.

37. prevention based questions? In the end

[SPMM]During your CAMHS rotation you become interested in strategies to prevent depression in school age young people. You decide to join a local project which is offering group training on coping strategies and mindfulness to all year 9 students, with additional 1:1 support for those who are identified to have risk factors for depression. This is an example which of the following types of prevention?

Select one:

A. Selective prevention
B. Universal prevention
C. Universal with indicated prevention
D. Indicated prevention
E. Universal with selective prevention

This intervention targets all students (universal) and then students who are deemed to be at higher risk for developing depression (selective). Universal interventions target the general public or a whole population group. Selective interventions focus on a particular group or subgroup who have biological, psychological or social risk factors for developing a mental disorder. Indicated prevention targets high-risk individuals with detectable symptoms but whom do not yet meet the criteria for the disorder.

38. Statistical tests??

[SPMM] Theme: Statistical Tests:
For each of the following vignettes choose the most appropriate statistical test/s.

A. ANOVA
B. ANCOVA
C. Chi-squared
D. Fisher's test.
E. Logistic regression.
F. Mann-Whitney test.
G. Paired t-test.
H. Unpaired t-test.

1. Comparison of the risk of developing schizophrenia in two groups of 3000 people - one group comprises of people who have used cannabis and the other of people who have never used cannabis.
2. Investigation of the association between several risk factors and the risk of developing schizophrenia.
3. Comparing baseline and endpoint weight in the same individuals in a phase II study of an antipsychotic. The variable is normally distributed. (Choose **TWO**)

Explanation:

Risk is a proportion; comparing proportions as in question 1 can be done using Chi-square test. In question 2, developing schizophrenia is a binary outcome (present or absent). Hence logistic regression is useful as a multivariate regression technique when several risk factors are evaluated. In question 3, two mean weight values are compared - but one measure is dependent on the other (paired). Hence either paired t test or repeated measures ANOVA can be used.
Correct Answer is : 1. Chi-squared. 2. Logistic regression. 3. ANOVA., Paired t-test.

https://www.youtube.com/watch?v=CnLO7wwt7n0&t=2s&ab_channel=AmourLearning

39. Reformulation, right?

[SPMM]Cognitive analytic therapy is structured into established steps, which the patient moves through over the course of the sessions. Which of the following occurs in the initial sessions of CAT?

Select one:

A. Recognition
B. Reformulation
C. Exploration
D. Exit
E. Revision

The initial sessions of CAT are devoted to gathering information about current difficulties and previous experiences, after which the therapist writes the patient a **reformulation** letter. This phase is followed by a **recognition** phase where the patient is helped via various means to acknowledge their difficulties and how they occur. This is followed by a **revision** phase where the patient and therapist plan together how the patient will cease unhelpful behaviours and patterns of thinking.

- **Initial phase:** an **exploration** of traps, dilemmas and snags. Therapist writes formulation letter
- **Middle phase:** working through problems with the use of diagrams exploring 'target problem procedures.'
- **Ending phase:** both patient and therapist write goodbye letters

40. There was a question regarding ethics of human research. Was the answer Belmont report or **Helsinki declaration**?

The Belmont Report was written by the National Commission for the Protection of Human Subjects of Biomedical and Behavioural Research. The Commission, created as a result of the National Research Act of 1974, was charged with identifying the **basic ethical principles** that should underlie the conduct of biomedical and behavioural research involving human subjects and developing guidelines to assure that such research is conducted in accordance with those principles. Informed by monthly discussions that spanned nearly four years and an intensive four days of deliberation in 1976, the Commission published the Belmont Report, which **identifies basic ethical principles** and guidelines that address ethical issues arising from the conduct of research with human subjects. *Ref: https://www.hhs.gov/ohrp/regulations-and-policy/belmont-report/index.html*

- **Nuremberg → Code of Ethics**
- **Declaration of Malta → Hunger strike,**
- **Declaration of Geneva → Revision of the Hippocratic oath,**
- **Tokyo → Not to participate in torture of prisoners or detainees,**
- **Lisbon → International statement of rights of pts,**
- **Ottawa → Principles for optimal child health**

41. First step in DBT after **pre-preparation**???

- The four **modes** of treatment in DBT are as follows: (1) group skills training, (2) individual therapy, (3) phone consultations, and (4) consultation team
- Key Techniques (mnemonic ***DICE***)
  - Distress tolerance includes accepting, finding meaning for, and tolerating distress. This includes crisis survival strategies such as distracting, self-soothing, improving the moment, and thinking of pros and cons and acceptance skills such as radical acceptance, turning the mind toward acceptance, and willingness versus willfulness.
  - Interpersonal effectiveness training is very similar to assertiveness and problem-solving training.
  - Core mindfulness training - learning to monitor internal mental states.
  - Emotion regulation skills form an important part of DBT.
- **DBT also involves social skills training** such as meditation, assertiveness training, etc. Another approach commonly employed in DBT is v**alidation** - recognizing distress and behaviours as legitimate and understandable but ultimately harmful.

reducing the quality of life.

42. Self-harm - It said something like "including people who are detained and are unable to consent".

43. **Bracketing and member checking**

[SPMM] Expectations and preconceptions can potentially taint the process of qualitative research. Which of the following methods is used in qualitative research to mitigate the potentially deleterious effects of researcher preconceptions?

Select one:

A. Publishing the protocol
B. Triangulation
C. Bracketing
D. Focus groups
E. Pre-registering the study

Bracketing: [Psych-Mentor]

Bracketing is a methodological device of phenomenological inquiry that requires **deliberate putting aside one's own belief about the phenomenon under investigation or what one already knows about the subject prior to and throughout the phenomenological investigation.**

- ***Respondent validation**: ("**member checking**") This includes techniques in which the investigator's account is compared with those of the research subjects to establish the level of correspondence between the two sets.*

44. human rights in research is **Helsinki** or Nuremberg , Helsinki was the available option

Declaration of Helsinki 1964: This was adopted by The 18th World Medical Association General Assembly in 1964 and has been amended five times since, most recently in 2000. Notes of clarification were added in 2002 and 2004. The current (2004) version is the only official one. The Declaration specifically addresses clinical research, reflecting changes in medical practice from the term 'Human Experimentation' used in the Nuremberg Code.

Nuremberg Code 1974: Code of ethics following the Nuremberg Trials (post-World War II Trial concerning doctors experimenting on people detained in concentration camps). According to Nuremberg Code, human experimentation can be carried out only if:

- Voluntary consent is given
- Research is intended for common good of the society
- Avoidance of unnecessary pain and suffering is guaranteed for the subjects
- Subject has liberty to withdraw at any point
- Qualified researchers undertake research
- Scientist must terminate a study if more harm is being caused than expected to the subjects

45. Test to check for heterogeneity - **Cochrane Q?**

- Cronbach alpha
- Kappa – inter-rater reliability.
- Forest plot
- Galbraith plot

- **Nominal data** use Kappa
- **Ordinal data** use **weighted kappa**

Internal consistency: is the extent to which items on a test measure various aspects of the same characteristic and nothing else.

There are four main ways to assess it:-

- Average inter-item correlation
- Average item-total correlation
- Split half correlation
- Cronbach's alpha

[SPMM]In a meta-analysis, what statistical measure is commonly used to assess the degree of heterogeneity among included studies?

Select one:

A. I-squared (I^2) Statistic
B. T-score

C. Mean Effect Score
D. Chi-squared Test for Independence
E. Standard Error.

EXPLANATION: In meta-analytical studies, the I statistic is a widely used measure to assess the degree of heterogeneity among included studies. It quantifies the proportion of total variation across studies that is due to heterogeneity rather than chance. I values typically range from 0% to 100%. Higher I values indicate greater heterogeneity. This statistic is essential for understanding the variability in effect sizes among studies and plays a key role in determining whether a random-effects or fixed-effects model is more appropriate for synthesizing study findings.

46. Valproate associated fever/abd pain **pancreatitis**

[SPMM] A 45-year-old patient with bipolar I disorder has recently started taking a new medication. He develops sudden and severe epigastric painthat radiates to his back, as well as vomiting. Which of the following medications is most likely to be responsible?

Select one:

A. Sodium valproate
B. Lithium
C. Lamotrigine
D. Carbamazepine
E. Topiramate

Pancreatitis presents as sudden and severe epigastric pain that radiates to the back and is often associated with vomiting. Systemicinfl ammation results in temperature spikes and patients can rapidly lose intravascular volume requiring aggressive fl uid resuscitation.Though the most common causes of pancreatitis are gallstones and alcohol, drug-induced pancreatitis has been associated withantipsychotic and mood stabiliser use, in particular sodium valproate.

[SPMM] A 39-year-old woman has a history of bipolar affective disorder and is on sodium valproate 1000mg/day. She has developed nausea and abdominal pain radiating to the back.

Explanation: Sodium valproate can cause both **gastric irritation and hyperammonaemia**, both of which can lead to nausea. It can also cause **pancreatitis**, during which the patient can present with abdominal pain radiating to the back. Investigation of choice is **CT abdomen.**

47. Hypercalcemia in a bipolar patient:

A. **Lithium**.
B. Carbamazepine.
C. Topiramate.
D. Valproate.

[SPMM] 52-year-old patient on maintenance treatment for bipolar I disorder is noted to have hypercalcemia on routine blood work. Which of the following agents is most likely to be responsible?

Select one:

A. Lithium
B. Valproate
C. Topiramate
D. Carbamazepine
E. Haloperidol

Barbiturates, carbamazepine, haloperidol, and valproate have all been reported to lower calcium levels, while lithium has been reported toraise them. Lithium increases the risk of hyperparathyroidism, and some recommend that calcium levels should be monitored in patients on long-term treatment. Clinical consequences of chronically increased serum calcium levels include renal stones, osteoporosis, dyspepsia, hypertension, and renal impairment

Reversible hyperparathyroidism and hypercalcaemia are associated with lithium therapy. Lithium also causes **hyperthyroidism**

Most adverse effects of lithium are **dose and plasma level related**, including mild GI upset, fine tremor, polyuria, and polydipsia. Polyuria may occur more frequently with twice-daily dosing. **Tremor**, primarily of the hands, is among the **most common lithium side effects**, seen in approximately 25% of treated patients. Lithium increases the risk of **hypothyroidism**; in middle-aged women, the risk may be up to **20%**. Lithium also more rarely increases the risk of hyperthyroidism and hyperparathyroidism. Lithium can also cause a **metallic taste in the mouth**, ankle oedema, and weight gain. Lithium causes less weight gain than valproate.

- **Hypocalcaemia – DiGeorge syndrome**
- **Disulfiram – metallic taste**

48. **Narrative analysis**? Not sure which narrative but possible stats question.

- *Narrative review in stats*

	Narrative review	Systematic Review
Question	Broad in scope; general and vague sometimes	Focused and narrow
Search methods	Not usually specified; comes from an experts familiarity with the literature	Comprehensive & explicit; using well defined search strategies
Appraisal of individual studies	Variable; usually sails on the expert's opinion	Rigorous appraisal is made to each individual study included irrespective of the direction of the outcome
Synthesis of results	Often provides only a qualitative summary of studies	Quantitative summary (using meta-analysis) is possible
Resources	Comparatively less time consuming; but requires an expert in the field to guide the process	Highly time consuming; no need for field expertise; in fact non experts can carry out an unbiased search than the experts at times.

[SPMM] An old age psychiatrist is interested in reviewing the literature on hyponatraemia related to SSRI use. He has wide clinical experience on this topic and chooses some key review papers as starting point and collates the information known to him already to provide a summary of evidence. Which research method is he following?

Select one:

A. Meta-analysis
B. Qualitative research
C. Case series
D. Systematic review
E. **Narrative review**

Narrative reviews are usually carried out by experts in the field of study and are more or less close to an expert's opinion supported by research evidence. These reviews are generally broad without a narrow or specific clinical question and are subjective. They carry a higher risk of being prone to bias than systematic reviews. Nevertheless narrative reviews can provide a good introduction to a topic, stimulate interest and controversies and can even collate existing data for generating new hypotheses.

*N.B : **Narrative family therapy – key words**: externalisation and unique outcomes (therapist highlight outcomes) – difficulties are a reflection of unhelpful dominant problem saturated narratives*

49. First line antidepressant in 13yr old.
Fluoxetine start from 5 years old

50. Phenomenological?

Phenomenology: [Psych-Mentor]

Phenomenology is a **qualitative research** approach that seeks to understand the subjective experience of individuals by exploring the meaning they attach to their lived experiences. It involves the study of how individuals experience and make sense of the world around them, without imposing any preconceived notions or theories.

Phenomenological research is typically conducted through in-depth interviews with participants who have experienced a particular phenomenon of interest. The researcher uses open-ended questions to encourage participants to describe their experiences in as much detail as possible. The goal is to gain a deep understanding of how the participants experience the phenomenon, including their thoughts, feelings, and perceptions.
The data collected through interviews is analysed using a process called 'bracketing.' This involves setting aside any preconceived notions or biases that the researcher may have and analysing the data objectively. The researcher looks for patterns and themes in the data, and then identifies the essential features of the phenomenon being studied.

51. There's was one about LBD and psychotic Sx - **rivastigmine.**

[SPMM] What is the most common medication used for treating patients with Lewy body dementia and psychosis?

Select one:

A. Fluoxetine
B. Haloperidol
C. Memantine
D. Rivastigmine
E. Risperidone

Explanation: The first line management of cognitive impairment, psychosis and agitation of Lewy body dementia (DLB) and Parkinson's disease dementia (PDD) is a cholinesterase inhibitor (ChEI). In a large RCT, **Rivastigmine** alleviated visual hallucinations, delusions, anxiety and apathy in DLB (McKeith et al 2006). Levodopa induced hallucinations are also less common when ChEIs are co-prescribed. There is a risk of neuroleptic sensitivity even with atypical antipsychotics of which there are no placebo-controlled trials in DLB. **Rivastigmine is a drug of choice when there is co-morbidities**

And TD in someone on long term FGA and procyclidine. Reduce dose of tetrabenazine seemed like possible answers. [There were 2, one was hallucinations and the other was psychosis there was a third Q in the stem of LBD too. [It was reported that the question was a scenario about LBD and different side effect and ask to choose medication for each one.]
I1) **rivastigmine and 2) reduce L-dopa,** third Sx have got worse after a medication, what caused it?
And one of the options was L dopa
And 3rd was what to do when somebody on cocarel

the following table from the Maudsley Guidelines provides a good summary of EPSEs

	Dystonia	Pseudo-parkinsonism (e.g. tremor)	Akathisia	Tardive dyskinesia
Prevalence (with older drugs)	Approximately 10% But more common in - young males - neuroleptic-naïve - high potency drugs (e.g. haloperidol)	Approximately 20% But more common in - elderly females - those with pre-existing neuro damage (e.g. stroke)	Approximately 25%	5% of patients per year of antipsychotic exposure But more common in - elderly women - those with affective illness - those who have had EPSE early on in treatment
Time taken to develop	Acute dystonia can develop within minutes or hours of starting antipsychotics	Days to weeks after antipsychotic started or dose increased	Hours to weeks	Months to years
Treatment	• Anticholinergic drugs* • Switching to alternative antipsychotic • Botulinum toxin	• Reduce the dose • Anticholinergic drugs***** • Switching to alternative antipsychotic	• Reduce the dose • Switching to alternative antipsychotic (quetiapine/ olanzapine) • Propranolol • **Mirtazapine** or **mianserin** (low dose) • Anticholinergic drugs* • Cyproheptadine • Benzodiazepine (e.g. diazepine or clonazepam) • Clonidine	• Stop anticholinergic (if prescribed) • Reduce the dose • Switch to an atypical (clozapine and quetiapine have best evidence) • Tetrabenazine • Ginkgo biloba

***Anticholinergics** used include **trihexyphenidyl, procyclidine, orphenadrine, benztropine.** The antihistamine, **diphenhydramine**, is also used due to its anticholinergic properties. The extrapyramidal side effects (EPSE's) are a group of side effects that affect voluntary motor control. They are most commonly seen in patients taking antipsychotic drugs but have also been reported in the use of SSRI's and tricyclics (TCA). Higher rates are also seen in patients with schizophrenia who have never received medication suggesting EPSE may be a feature of the illness itself (McCreadie, 2002).

Q. from another recall, Tardive dyskinesia, on flupentixol and procyclidine for 20 yrs., what do you do?

A. **Stop procyclidine**
B. Start tetrabenazine
C. Decrease flupentixol dose
D. Something else, can't remember

52. Non-dominant parietal stroke- was it **anosognosia** or dyscalculia?

Other Symptoms: contralateral **hemi-spatial neglect** (hemineglect), dressing apraxia and construction apraxia and anosognosia.

➲ *A specific syndrome, **Gerstman's syndrome** is associated with lesions of the **DOMINANT PARIETAL LOBE**. This consists of **dyscalculia, agraphia, finger agnosia and right-left disorientation**.*

53. PD vignette EMIs. ICD-11 PD with...? detachment, anankastic and borderline (or disinhibited, wasn't sure)

Note: 'Multiple personality disorder' is a dissociative / conversion disorder and is not classified within the personality disorder sections. In the DSM-5 it is referred to as 'Dissociative Identity Disorder'.

ICD-11 trait domain	Description
Negative Affectivity	The core feature of the Negative Affectivity trait domain is the tendency to experience a broad range of negative emotions. Common manifestations of Negative Affectivity, not all of which may be present in a given individual at a given time, include: experiencing a broad range of negative emotions with a frequency and intensity out of proportion to the situation; emotional lability and poor emotion regulation; negativistic attitudes; low self-esteem and self-confidence; and mistrustfulness.
Detachment	The core feature of the Detachment trait domain is the tendency to maintain interpersonal distance (social detachment) and emotional distance (emotional detachment). Common manifestations of Detachment, not all of which may be present in a given individual at a given time, include: social detachment (avoidance of social interactions, lack of friendships, and avoidance of intimacy); and emotional detachment (reserve, aloofness, and limited emotional expression and experience).
Dissociality	The core feature of the Dissociality trait domain is disregard for the rights and feelings of others, encompassing both self-centeredness and lack of empathy. Common manifestations of Dissociality, not all of which may be present in a given individual at a given time, include: self-centeredness (e.g., **sense of entitlement**, expectation of others' admiration, positive or negative attention-seeking behaviours, concern with one's own needs, desires and comfort and not those of others); and **lack of empathy** (i.e., indifference to whether one's actions inconvenience hurt others, which may include being deceptive, manipulative, and exploitative of others, being mean and physically aggressive, callousness in response to others' suffering, and ruthlessness in obtaining one's goals).
Disinhibition	The core feature of the Disinhibition trait domain is the tendency to act rashly based on immediate external or internal stimuli (i.e., sensations, emotions, thoughts), without consideration of potential negative consequences. Common manifestations of Disinhibition, not all of which may be present in a given individual at a given time, include: impulsivity; distractibility; irresponsibility; recklessness; and lack of planning.
Anankastia	The core feature of the Anankastia trait domain is a narrow focus on one's rigid standard of perfection and of right and wrong, and on controlling one's own and others' behaviour and controlling situations to ensure conformity to these standards. Common manifestations of Anankastia, not all of which may be present in a given individual at a given time, include: perfectionism (e.g., concern with social rules, obligations, and norms of right and wrong, scrupulous attention to detail, rigid, systematic, day-to-day routines, hyper-scheduling and planfulness, emphasis on organisation, orderliness, and neatness); and emotional and behavioural constraint (e.g., rigid control over emotional expression, stubbornness and inflexibility, risk-avoidance, perseveration, and deliberativeness).

54. antipsychotic in diabetic retinopathy? Indicted

A. **Amisulpride**
B. Haloperidol
C. Risperdal

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4838103/

Summary: Antipsychotics – risk of diabetes and impaired glucose tolerance

High risk	Clozapine, olanzapine
Moderate risk	Quetiapine, risperidone, phenothiazines
Low risk	High-potency FGAs (e.g. haloperidol)
Minimal risk	Aripiprazole, amisulpride, asenapine, brexpiprazole, cariprazine, lumateperone, lurasidone, ziprasidone

N.B: Topiramate contraindicated in angle closure glaucoma

55. Korsakoff test by digit span or address test; Digit span is normal; **address recall** should be the answer

Patient with Korsakoff syndrome:

1. Overall WAIS is low due to frontal lobe dysfunction (although digit span is conserved)
2. Episodic memory, and hence MMSE and AMT will be affected.
3. MMPI score is high due to frontal lobe damage.

- Patient with Korsakoff **maintains procedural, attention and working memory** (these three are unaffected in Korsakoff)

56. Least psychologically addictive: **LSD**

57. Med to assist with withdrawal in alcohol dependence but the guy takes tramadol for back pain and maybe has hypotension.

A. **Acamprosate**,
B. Naltrexone.
C. Disulfiram

[SPMM] A 40-year-old woman has completed detoxification for alcohol. She is interested in taking some medication to help her maintain abstinence. She is on tramadol for chronic back pain and has a history of ischaemic heart disease. Which of the following medications would be the best choice for this patient?

Select one:

A. Acamprosate
B. Nalmefene
C. Naltrexone
D. Diazepam
E. Disulfiram

Explanation: Acamprosate is licensed for the treatment of alcohol dependence in the UK and may be offered in combination with psychosocial treatment. It is indicated for patients who want to remain abstinent from alcohol. Both naltrexone and nalmefene are opioid antagonists and should not be given together with opioid agonist drugs (such as tramadol). There is no place for continued use of benzodiazepines (such as diazepam) beyond the treatment of an acute alcohol withdrawal syndrome, and benzodiazepines are not indicated for relapse prevention. **Disulfiram is contraindicated in the setting of coronary artery disease**

58. Which medication reduces seizure duration in ECT - **Chlordiazepoxide**

The table below summarises the effect of the important psychotropics and their effect on seizure duration.

Psychotropic class	Effect on seizure duration	Advice
Benzodiazepine	Reduced ↓	Avoid where possible
SSRIs	Minimal effect	
Venlafaxine	Minimal effect	
TCAs	Possibly increased ↑	TCAs are associated with arrthymia following ECT in the elderly and those with cardiac disease so should be avoided in ECT in these groups

Psychotropic class	Effect on seizure duration	Advice
MAOIs	Minimal effect	
Lithium	Possibly increased ↑	Generally used in ECT without significant problems
Antipsychotics	Some potential increase in clozapine and phenothiazines, other antipsychotics considered ok ↑	Limited data
Anticonvulsants	Reduced ↓	If used as mood stabiliser continue but be prepared to use higher energy stimulus

59. What was the answer for the question about differentiating autism and schizophrenia?
    Some options were:
    A. Mood incongruent symptoms
    B. Catatonia,
    C. Normal or near normal development.

    See Q31.

    [SPMM] Please match the scenarios described with the most likely cause for catatonia.

    Choose...
    Neuroleptic Malignant Syndrome
    Depressive Stupor
    Manic stupor
    Autism
    Schizophrenia
    Postencephalitic state
    Intracranial space-occupying lesion

    A 42-year old man is changed from an oral antipsychotic, to a depot injection. A week after the depot he complains of stiffness in his arms, and the following day he lies rigidly in bed, sweating profusely, with his eyes moving around and not able to rest on your face as you attempt to talk to him

    A 17-year old young man has been brought onto the ward after an aggressive outburst at home. He is now standing very still, with his torso leaning forwards, and has kept this position for two hours. You try to engage him in conversation, and his father tells you that he does not communicate verbally, and never has done

    A 22 -year old woman is brought to hospital by her friends after she stopped moving several hours previously. They tell you that for the past three days she has been out of the house all day and all night, that she reported feeling completely wonderful, and had spent all her savings on unusual jewellery

    Explanation:

    Q1: This man is now presenting with features of Neuroleptic malignant syndrome, which includes muscular rigidity, hyperthermia and diaphoresis. Administering a depot injection is considered as one of the risk factors for developing NMS.

    Q2: A young man who does not have verbal speech and who has had an aggressive outburst may suggest he has Autism. People with Autism can sometimes enter catatonic states or maintain postures for long periods of time.

    Q3: The history preceding the stupor is suggestive of mania (elated mood, excessive spending, lack of sleep/increased energy).

60. Person admitted with poly substance use, 2 days later had respiratory depression with use of accessory muscles, tachycardia, hypotension. what medication to use.

    A. **Naloxone**,

B. Naltrexone,
C. Buprenorphine, etc

We think opioid toxicity so Naloxone should be the answer, however 2 days following admission is a bit confusing and sounds like withdrawal.

Drug	UK Street names	Effects	Withdrawal features
Opioids	Heroin = Smack, horse, junk, skag, brown, Harry, H	Euphoria, drowsiness, constipation, pupillary constriction, respiratory depression	Piloerection, insomnia, restlessness, dilated pupils, yawning, sweating, abdominal cramps
Amphetamine and cocaine	Amphetamines = Speed, meth, ice, uppers, whizz. Cocaine = Snow, crack, coke, rock	Increased energy, insomnia, hyperactivity, euphoria, paranoia, reduced appetite	Hypersomnia, hyperphagia, depression, irritability, agitation, vivid dreams, increased appetite
MDMA (ecstasy)	MDMA = XTC, E, Eccies, pills, Adam, Mandy	Increased energy, increased sweating, jaw clenching, euphoria, enhanced sociability, increased response to touch	Depression, insomnia, depersonalisation, derealisation
Cannabis	Cannabis = Marijuana, weed, pot, grass, joint, mull, cone, ganja, hash, chronic	Relaxation, intensified sensory experience, paranoia, anxiety, injected conjunctivae	Insomnia, reduced appetite, irritability
Hallucinogens	LSD = Trips, Tripper, Tab, Stars, Smilies, Rainbows, Paper	Perceptual changes, pupillary dilation, tachycardia, sweating, palpitations, tremors, inco-ordination	No recognised withdrawal syndrome
Ketamine	Ketamine = Vitamin K, Super K, Special K, K, donkey dust	Euphoria, dissociation, ataxia, hallucinations, muscle rigidity	No recognised withdrawal syndrome

61. What increases schizophrenia risk?

A. Maternal cannabis use
B. Double bind situations as a child

Risk factors [Psych-Mentor]

- **Social class:** people of lower socioeconomic class are more likely to suffer from the condition.
- **Learning disability:** Prevalence rates for schizophrenia in people with learning disabilities are approximately three times greater than for the general population.
- **Race and ethnicity:** Early studies suggested schizophrenia was more common in black populations than in white. The current consensus is that there are **no differences** in the rates of schizophrenia by race and that differences that are found are largely attributable to ethnic minority status
- **Gender and age:** The consensus view is that schizophrenia is equally common in males and females, again there is evidence to the contrary suggesting it is more common in females.
- **Marital status:** Females with schizophrenia are more likely to marry than males.
- **Family history:**

Relationship to person with schizophrenia	Risk of developing schizophrenia
General population	1%

Relationship to person with schizophrenia	Risk of developing schizophrenia
First cousin	2%
Grandchildren	5%
Parents	6%
Siblings	9%
Children	13%
Fraternal twins	17%
Identical twins	48%

- **Season of birth:** The proportion of people with schizophrenia born in winter months is 5-15% higher than at other times of the year.
- **Urban versus rural place of birth:** The rates of schizophrenia have been shown to be higher in those born and/or raised in **urban areas.**
- **Obstetric complications.**
- **Migration:**

Three risk factors with good evidence for increasing the risk of schizophrenia are: [Psych-Mentor]

- **Prenatal nutritional deprivation**
- **Prenatal brain injury**
- **Prenatal influenza**

62. Commonest cause of mild LD.

A. Prenatal? Issues,
B. Perinatal injuries,
C. Genetics
D. Socioeconomic status and
E. Unknown cause

Note: that **Fragile X syndrome** is the most common **inherited** cause. The most common **environmental** cause is the **foetal alcohol syndrome.** Overall the **commonest** known cause is **Down's syndrome.** [SPMM]

[Psych-mentor] ID can be caused by environmental and/or genetic factors. However, for **up to 60% of cases, there is no identifiable cause** Genetic causes of ID are thought to be present in 25-50% of cases, although this number increases proportionally with severity. Down Syndrome is the most common chromosomal cause of ID. Fragile X syndrome is the most common genetic cause The most common acquired form of ID is foetal alcohol syndrome.

- Most Common **Inherited Cause** → **Fragile X Syndrome**
- Most Common **Genetic Cause** → **Down Syndrome Then Fragile X Syndrome**
- Most Common **Environmental Cause** → **Alcohol Syndrome**
- Most Common **Overall** → **Unknown.**

63. Question about drug induced Parkinsonism and Parkinson's disease (can't remember much)

A. Micrographia
B. Shuffled gait
C. Become worse with time.
D. Symmetrical

Gait	Description	Associated conditions
Festinating (Shuffling)	Posture is stooped forward. Gait initiation is slow and steps are small and shuffling	Parkinson's disease
Ataxic	Gait is wide-based with truncal instability and irregular lurching steps which results in lateral veering and if severe, falling	Cerebellar disease, e.g. Wernicke's
Antalgic	Stance phase of gait is abnormally shortened relative to the swing phase, usually done to minimise pain	Lower limb trauma
Spastic (scissor or diplegic)	Rigidity and excessive adduction of the leg in swing , plantar flexion of the ankle, flexion at the knee, adduction and internal rotation at the hip, and contractures of all spastic muscles	Cerebral palsy
Steppage (neuropathic or equine)	High-stepping gait so as to prevent scraping of the toe on the ground	Foot drop
Myopathic (waddling)	A broad-based gait with a duck-like waddle to the swing phase, the pelvis drops to the side of the leg being raised with forward curvature of the lumbar spine, and a marked body swing	Proximal myopathy
Pigeon	Toe(s) point(s) inwards when walking	Structural abnormalities
Trendelenburg	During the stance phase, the weakened abductor muscles allow the pelvis to tilt down on the opposite side. To compensate, the trunk lurches to the weakened side to attempt to maintain a level pelvis throughout the gait cycle	Poliomyelitis or muscular dystrophy
Stomping	Bilateral high steppage due to lack of proprioception	Friedreich's ataxia
Magnetic	Feet seem as if magnetically attracted to the floor	Normal pressure hydrocephalus
Choreiform (hyperkinetic)	Irregular, jerky, involuntary movements in all extremities. Walking may accentuate their baseline movement disorder	Sydenham chorea, Huntington's
Sensory	In an effort to know when the feet land and its location, the patient will slam the foot hard onto the ground in order to sense it	proprioceptive loss e.g. diabetes, B12 deficiency

64. Was the case of old lady who struggled with finances but lived independently **MCI**?

65. Children with social anxiety, increased risk in adult life for …?

A. **Substance misuse**
B. Schizoid
C. Schizotypal
D. Borderline

66. How to differentiate MCI from Dementia. I chose informant assessment of ADLs but not sure at all, probably not correct

A. MoCA
B. MRI,
C. Informant assessment of memory,
D. **Informant assessment of ADLs**

☞ **The Montreal Cognitive Assessment (MoCA)** is a test used to detect mild cognitive decline and early signs of dementia. It can help identify people at risk of Alzheimer's disease and screen for conditions like Parkinson's disease, brain tumors, substance abuse, and head trauma.

Introduced in 2005, the MoCA test is an update from the older Mini-Mental State Examination (MMSE) introduced in 1975. It contains 30 questions and takes 10 minutes to complete. While the MOCA test is useful in detecting dementia, it cannot differentiate between the different dementia types. It is available in more than 35 languages as well as **versions for people with blindness or hearing impairment.**

[Psych-Mentor] **There are two clinical screening instruments which allow for a broad assessment of mild cognitive impairment the CAMCog (part of the CAMDEX) and the SISCO (part of the SIDAM).**

[Psych-Mentor] **Which of the following is true regarding mild cognitive impairment (MCI)?**

A. **MCI represents a middle ground between normality and dementia**
B. The principal cognitive impairment is always amnestic
C. People with mild cognitive impairment never recover
D. To meet a diagnosis of MCI a patient must demonstrate impairment in basic activities of daily living
E. MCI is a mild form of dementia

Correct answer : MCI represents a middle ground between normality and dementia. Mild cognitive impairment defines a transitional stage between normal ageing and dementia.

67. EMI therapist 3 questions

A. Reframing
B. Confrontation??
C. Clarification

1. Therapist says 'you told me about an interview for a job, we've been talking for about 20 mins now and you have not mentioned anything about it. How did the interview go? **Reframing** *(We think answer is **confrontation**)*

2. Therapist says, 'you told me your mum is always out at night, what does she do when she goes out like that'. **Clarification**

68. When there is significant difference between dropout rate in a placebo and mirtazapine group, what is important in the study design to look out for.

A. Blinding,
B. Intention to treat analysis,
C. Sensitivity analysis,
D. Confounding factors

ITT analysis aims to reduce the impact of dropouts and attrition in prospective trials.

69. There was question in referring to AN. Most common abnormality on refeeding. Won't that be hyperkalaemia? **Hypophosphatemia**

[SPMM]The most likely electrolyte abnormality of refeeding a grossly anorexic patient in a medical ward is

A. Hypophosphatemia
B. Hyperkalemia
C. Hypercalcemia
D. Hypermagnesaemia
E. Hyponatremia

Excerpt from MARSIPAN report "In an acute medical service (Gaudiani et al, 2012) **45% of patients developed hypophosphatemia** and in French intensive care units (Vignaud et al, 2010) 11% of patients developed re-feeding syndrome. It appears, therefore, that the highest rate of re-feeding syndrome is reported from the units in which patients are most unwell, namely adult medical and intensive care setting". As noted in multiple sources, including MARSIPAN (Management of Really Sick Patients with Anorexia Nervosa), the potentially fatal refeeding syndrome is caused by the switch from fasting gluconeogenesis to carbohydrate-induced insulin release, **inducing considerable intracellular uptake of potassium, phosphate, and magnesium from the bloodstream. This can cause profound biochemical derangements including hypokalaemia, hypophosphatemia, hypomagnesaemia, and occasionally hypocalcaemia**

70. How to ensure almost same sample size for a study? **Block randomisation**, stratification

[SPMM] A researcher wants to make sure that the size of the control and two intervention groups are more or less equal so that statistical deductions can be meaningful. Which of the following methods of randomisation could ensure this?

Select one:

A. Block randomisation
B. Stratification
C. Cluster randomisation
D. Quasi-randomisation
E. Simple randomisation

Explanation: Block randomization refers to a method where randomization **occurs in a set of specified numbers, say blocks of 6 etc., to ensure equal number distribution between two arms**. Larger the block size, lesser the ability to predict allocation.

In **cluster randomization**, the unit of randomization and analysis is various centres or catchment-zones and not individual patients.
Stratification is classifying to subgroups before randomising within each subgroup to ensure impartial allocation of subgroup factors.

Quasi-randomisation refers to 'randomizing' using even/odd numbers of the date of birth, day of the week patient was seen, etc.

[Psych-Mentor] Which of the following aims to address confounding factors in the analysis stage of a study?

A. Randomization
B. Stratification
C. Meta analysis
D. Concealed allocation
E. Blinding

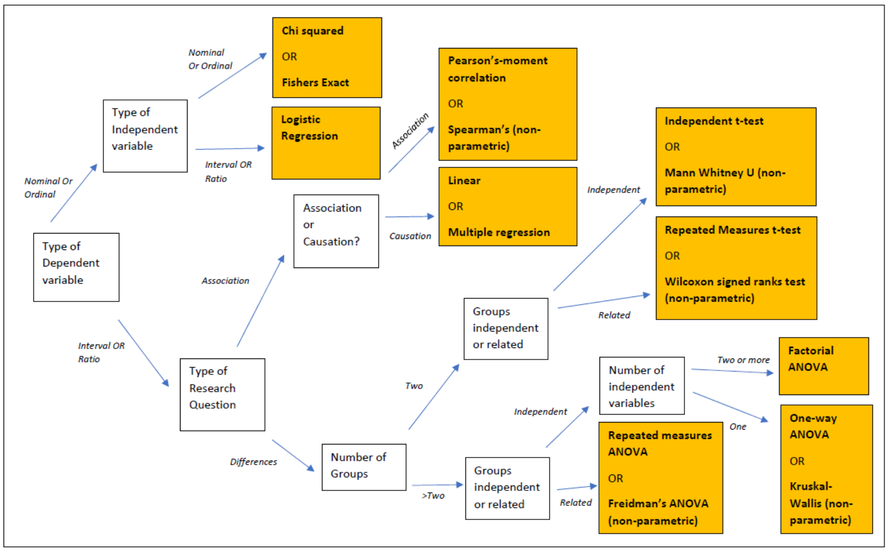
CORRECT ANSWER : Stratification Stats Confounding A confounding factor is an additional factor that obscures the relationship between an exposure and the outcome. Confounding factors are associated with both the exposure and the disease. Consider a study which finds that people who drink coffee are more likely to develop heart disease. The confounding factor in this study is smoking. Smoking is associated with both drinking coffee and heart disease. People who drink coffee are also more likely to smoke. In this case smoking confounds the apparent relationship between coffee and heart disease. Confounding occurs when there is a non random distribution of risk factors in the populations. Age, sex and social class are common causes of confounding. In the design stage of an experiment, confounding can be controlled by randomisation which aims to produce an even amount of potential risk factors in two populations. In the analysis stage of an experiment, confounding can be controlled for by stratification.

Q. Which of the following aims to address confounding factors in the design **stage** of a study?

A. Stratification

B. Bias sampling
C. Minimisation Funnel plots
D. **Randomisation**

71. Questions about statistical tests.



Different Statistical Tests

	Comparison				Association	Regression
	2 data sets		>2 data sets		Correlation between 2 variables	Predicting one from another
	Paired	Un-paired	Paired	Un-paired		
Normal Distribution (Mean)	Paired t test	Un-paired t test	Repeated measure ANOVA	One way ANOVA	Pearson correlation	Linear regression
Non - Normal Distribution (Median)	Wilcoxon signed-rank test	Wilcoxon rank sum test Mann-Whitney U test	Friedman test	Kruskal–Wallis test	Spears Rank correlation	Non Parametric Regression
Dichotomous data	McNemar's Test	Chi-square test Fisher's exact test	Cochran's Q test	Chi-square test	Contingency Coefficient	Logistic Regression

Choosing appropriate statistical tests

Lead in: For each study described below choose the most appropriate statistical test/s from the list above.

Three groups of randomly assigned patients with dementia were followed up for 2 years to compare the efficacy of three different cholinesterase inhibitors. Choose the most appropriate statistical method to compare mean change in their cognitive scores. **(Choose ONE):** ✖

Serum triglyceride levels of two groups of patients - one taking clozapine as inpatients with monitored diet and exercise program and the other taking clozapine as outpatients without a diet or exercise program - were evaluated. The mean triglyceride levels at the end of the period of observation in both groups were compared with each other. **(Choose TWO):** ✖ ✖

A control patient of the same age, race, and sex was found for each unmedicated depressed patient to make two groups of 25 each. The control patients were given a standard hypnotic and the test patients were given a new hypnotic to determine the difference in the number of hours slept during one night of observation. **(Choose THREE):** ✖ ✖ ✖

Regression analysis
F statistics (ANOVA)
Correlation analysis
Unpaired t test
Chi square test
Paired t test
One sample t test
Sign test

Explanation: Mean comparisons across more than two groups can be carried out using ANOVA. The dependent t-test for paired samples is used when the samples are paired. This implies that each individual observation of one sample has a unique corresponding member in the other sample. In case 3, as the subjects are matched, paired t test may offer a more robust test than the usual unpaired t test or ANOVA, though all three can be used.

The correct answer is: 1. F statistics (ANOVA)2F statistics (ANOVA) , Unpaired t test3F statistics (ANOVA), Unpaired t test, paired t test

72. There was a psychotherapy one on... proposed hypotheses. Maybe systemic family therapy?

Family therapy first began in the 1950's. This was a major shift in thinking and people's problems began to be considered in the context of their environments. There are five main models of family therapy to be aware of:

- Structural
- Strategic
- Systemic
- Transgenerational
- Solution Focused

Structural (developed by Salvador Minuchin) SCHAB

The main assumption is that the family's structure is wrong. Structural therapy has clear ideas about what constitutes a healthy family system. It is one where there are clear boundaries and no coalitions. The work is in the here-and-now. Dysfunctional families are thought to be marked by impaired boundaries, inappropriate alignments, and power imbalances. **Key terms include: subsystems, hierarchy, boundaries, alliances and coalitions. Therapist Expert stance. Transactional.**

Strategic (associated with Jay Haley and Cloe Madanes)

Strategic therapy claims that difficulties in families arise due to distorted hierarchies. Dysfunctional families are believed to communicate in problematic repetitive patterns (vicious cycles) that kept them dysfunctional. These patterns arise as intended solutions by the family to some symptom. The intended solutions then became the problem because the family had either over or under responded to the symptom through their interactions. **Key terms include: task setting, and goal setting, paradoxical interventions.**

Systemic (associated with Mara Selvini-Palazolli aka 'The Milan model')

Milan-Systemic therapists see the family as a self-regulating system which controls itself according to rules formed over a period of time through a process of trial and error. They are interested in the rule-maintaining characteristics of communication and behaviours, and assume that the way to eliminate a symptom is to change the rules. An interview consists almost entirely of questioning of the family by the therapist. Questioning is a recursive and circular process, with each question building upon the family's response to previous questions. Emphasis is placed on exploring differences between family member's behaviours, emotional responses and their beliefs at differing points in time. **Key terms include: hypothesising, neutrality, positive connotation, paradox and counter-paradox, circular and interventive questioning, and the use of reflecting teams**. **Non-expert stance**

Transgenerational

Transgenerational family therapy aims to understand how families, across generations, develop patterns of behaving and responding to stress in ways that prevent health development and lead to problems. Seven interlocking concepts make up the theory

- Scale of differentiation
- Nuclear family emotional system
- Family projection processes
- Multi-generational transmission processes
- Sibling position profiles
- Emotional cut-off

- Triangles

Solution Focused therapy

Solution Focused Solution focused therapy emphasises solutions over problems. In fact, it sees the problem as being maintained by a focus on the problem to the exclusion of the solution behaviour which is already happening. The central philosophy of the model could be stated as:

1. If it ain't broke, don't fix it.
2. Once you know what works, do more of it.
3. If it doesn't work, don't do it again, do something different.

Psychodynamic-psychoanalytic family therapy
Psychoeducational family therapy
Milan systemic therapy
Narrative family therapy
Structural family therapy
Strategic family therapy
Solution-focused family therapy

Approaches to psychotherapy in youth mental health
From the list provided, choose the type of family therapy that is characterised by the description.

1. Paradoxical interventions are commonly used ✖
2. Introduced the concepts of 're-framing' the problem and hypothesising, as well as ideas about neutrality and curiosity; therapist takes more of a non-expert stance ✖
3. Challenges to the hierarchy within the family results in problems; therapist takes a directive, 'expert' stance ✖
4. Difficulties are a reflection of dominant, problem-saturated stories; therapist highlights 'unique outcomes' and 'externalisation' ✖
5. Current family functioning is grounded in the past; core concepts include splitting; therapist creates a holding environment and offers interpretations ✖
6. Clear rules govern optimal family organisation, with a focus on hierarchy, subsystems, and boundaries ✖

Explanation:

In the model of strategic family therapy, problems always arise because of difficulties with hierarchy within the family system. Rather than attempting to resolve this, the family is ambivalent about having the problem, as it provides some gain for them. Reflecting this idea, the therapist takes a strategic stance to overcome their resistance, such as using 'paradox' (e.g. suggesting the problem may not be resolvable).

In Milan systemic family therapy, there is an emphasis on family beliefs and meanings, and the idea of there being no single objective truth about the problem. The Milan model introduced the concepts of 're-framing' the problem and hypothesising, as well as ideas about neutrality and curiosity, with the therapist taking more of a 'not-knowing', non-expert stance.

In structural family therapy, Minuchin proposed that clear rules govern optimal family organisation and structure, with a focus on hierarchy, subsystems, and boundaries. Challenges to this structure result in problems which the family attempts to address. In this model, the therapist takes a directive, 'expert' stance.

In narrative family therapy, difficulties are a reflection of unhelpful, dominant, 'problem-saturated' narratives. The therapist helps to highlight 'unique outcomes' to challenge this narrative and, through the use of 'externalisation' (i.e. separating the problem from the person), helps to rewrite this to a more helpful one.

In psychodynamic-psychoanalytic family therapy, core concepts include object relations, projective identification, splitting, and scapegoating. Common strategies include creation of a holding environment, interpretation, and emphasis on therapeutic alliance. This type of family therapy probably influences family therapists' clinical work more than is usually acknowledged.

Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019. pp. 718-719.
Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 761-762.

The correct answer is: (1) Strategic family therapy; (2) Milan systemic family therapy; (3) Structural family therapy; (4) Narrative family therapy; (5) Psychodynamic-psychoanalytic family therapy; (6) Structural family therapy

73. Another recreational drug with BL paraesthesia and another Sx. I was stuck between **phencyclidine** and nitrous oxide,

People who have taken PCP often present with the classic toxidrome of PCP intoxication: [Psych-Mentor]

- **Violent behaviour**
- **Dysarthria**
- **Nystagmus**
- **Tachycardia**
- **Hypertension**
- **Anaesthesia**
- **Analgesia**

74. Child with 1 month Hx of movement like sexual something. **Copropraxia**, stereotyped something, can't remember other options.
I think there were motor and tourette's thrown in as options
Copropraxia? Since it's just 1 month

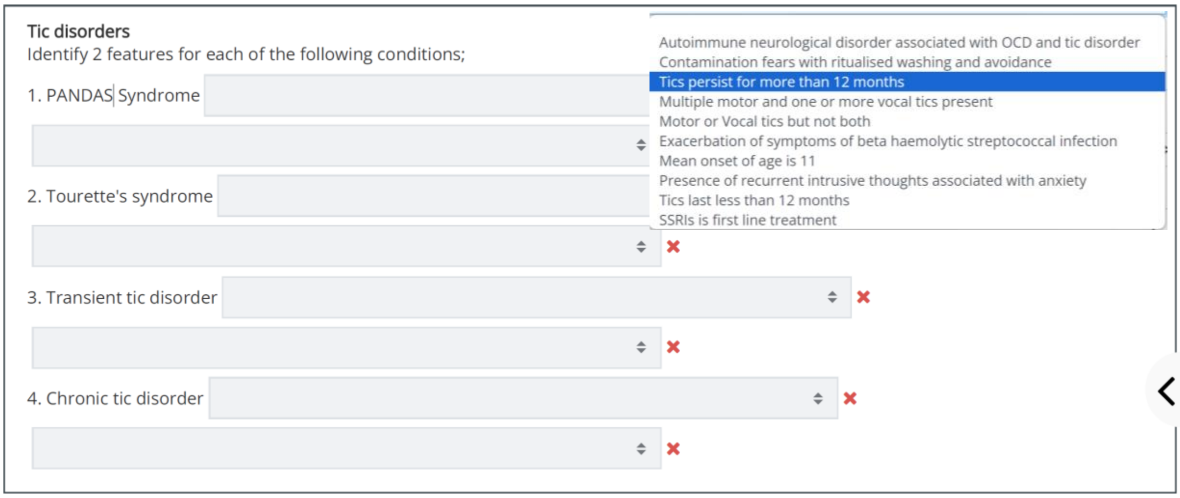
Copropraxia is a type of tic disorder characterized by involuntary and repetitive obscene or inappropriate gestures or movements. It is often associated with Tourette syndrome but can also occur in other neuropsychiatric conditions. In copropraxia, individuals may exhibit actions such as obscene gestures, inappropriate touching, or mimicking socially unacceptable behaviors. Treatment typically involves a combination of medication and behavioral therapy to manage symptoms and improve quality of life.

75. 8-year-old Pouting and facial grimacing for 6 months.

[Psych-Mentor] An 8 year old boy presents with a 6 month history of repeated eye blinking, grimacing, explosive use of obscene phrases and also obscene gestures. You decide to monitor his condition and see him again in 3 months. At the next visit the problems have resolved. What is the diagnosis at this stage?

A. TIC unspecified.
B. Tourette, chronic motor tic.
C. Chronic vocal tic.
D. Transient tic disorder.

- If there is 2 or more tics including vocal and motor tics for a year it's Tourette
ICD 11 says for primary tic disorder or chronic motor tic disorder symptoms have to be present for at least 1 year Tourette is Multiple Motor + 1 vocal


Tic disorders
Identify 2 features for each of the following conditions;
1. PANDAS Syndrome
2. Tourette's syndrome
3. Transient tic disorder
4. Chronic tic disorder

Autoimmune neurological disorder associated with OCD and tic disorder
Contamination fears with ritualised washing and avoidance
Tics persist for more than 12 months
Multiple motor and one or more vocal tics present
Motor or Vocal tics but not both
Exacerbation of symptoms of beta haemolytic streptococcal infection
Mean onset of age is 11
Presence of recurrent intrusive thoughts associated with anxiety
Tics last less than 12 months
SSRIs is first line treatment

Explanation:

1) PANDAS syndrome: Paediatric autoimmune neurological disorder associated with streptococcus. An autoimmune syndrome associated with OCD and tic disorder. Some children appear to develop obsessive-compulsive symptoms associated with beta haemolytic streptococcal infection and this presentation represents a small proportion of OCD cases in this population. This association has led to the studies of immune responses in OCD. Specific characteristics include onset before puberty, episodic or 'sawtooth' course and choreiform movements.

2) Tourette's syndrome is characterised by multiple motor and one or more vocal tics, causing distress and impaired function. The tics occur many times a day (usually in bouts) nearly every day or intermittently throughout a period of more than one year. During this period, there was never a tic-free period of more than three consecutive months. Facial tics: eye blinks or head jerks are often the initial symptoms, but tics involving the neck, shoulders, and upper extremities are also common.

3) Transient tic disorder: Single or multiple motor and/ or vocal tics (sudden, rapid, recurrent, nonrhythmic, stereotyped motor movements or vocalisations). Usually, face, nose, throat are involved (grimaces, blinks, frowns, grunts, throat clearing, sniffs). The tics occur many times a day, nearly every day for at least four weeks, but for no longer than 12 consecutive months.

4) Chronic Tic disorder persist for more than one year and often shows relapses and remissions throughout childhood. Maybe one or several tics are simultaneously present. Motor or Vocal tics but not both. Over time, one form fades to be replaced by another.

Correct Answer is : 1. Autoimmune neurological disorder associated with OCD and tic disorder, Exacerbation of symptoms of beta haemolytic streptococcal infection 2. Tics persist for more than 12 months, Multiple motor and one or more vocal tics present 3. Multiple motor and one or more vocal tics present, Tics last less than 12 months 4. Tics persist for more than 12 months, Motor or Vocal tics but not both

76. There was a question about an Adult-Shoplifting, What will prevent it from happening again?

A. Appearance in court.
B. Show remorse.

- **Only 3.2%** of the cases involved mentally ill patients
- There is a close link between shoplifting and affective disorders, alcoholism and drug addiction. **Depression is especially associated if the shoplifter is a middle-aged woman (24-30% middle aged female shoplifters may be depressed.**
- Shoplifting generally peaks in adolescence.
- **Most offenders do not reoffend if caught.**
- **Kleptomania** is an impulse control disorder. It is classified under ICD-10 F63 '**Habit and impulse disorders'.** In this case, the theft is usually petty and impulsive, done without the help of others or planning. Around 1-2% of shoplifters have a history consistent with kleptomania. Most kleptomaniacs are women. But the severity of kleptomania is not biased by gender. Nearly 50% of kleptomaniacs have a comorbid personality disorder. **The most common PD is paranoid personality followed by histrionic personality.**

[SPMM]Which of the following disorders is least likely to be associated with shoplifting?

Select one:

A. Obsessive-compulsive disorder
B. Schizophrenia
C. Depression
D. Substance misuse
E. Personality disorders

Burglary, theft, and fraud are common offences that are rarely associated with psychiatric disorder. Shoplifting has attracted some clinical attention. About 5% of shoplifters have a significant mental disorder (personality disorder, **substance misuse**, depression, schizophrenia, dementia). Pure kleptomania (compulsive stealing) is extremely rare. **Most common is substance misuse**

77. Suicide within 1 year of self-harm attempt. **0.7% (0.5 in females and 1.1 males)**

- **0.7% (0.5 in females and 1.1 males)**
- **Within 3 months of discharge from inpatient: 25 % and 40% before their scheduled follow up.**
- **Who visited GP in last one week: 40% (same 40% in older adults)**

- ➲ **More than 10 years after first attempt: 3%**
- ➲ **Life time prevalence of suicide : 4%**
- ➲ **Life time prevalence of suicidal thoughts: 15%**

78. Question on red flag MEED (eating disorders): HR 50bpm, **severe dehydration**??
    The question was asking which 1 is life threatening
    A. HR 50 bpm? (amber)
    B. Hypoalbuminemia
    C. Sever dehydration

	Red: High impending risk to life	Amber: Alert to high concern for impending risk to life	Green: Low impending risk to life
Medical history and examination			
Weight loss	Recent loss of weight of ≥1kg/week for 2 weeks (consecutive) in an undernourished patient[34] Rapid weight loss at any weight, e.g. in obesity or ARFID	Recent loss of weight of 500–999g/week for 2 consecutive weeks in an undernourished patient[126]	Recent weight loss of <500g/week or fluctuating weight
BMI and weight	• Under 18 years: m%BMI[35] <70% • Over 18: BMI <13	• Under 18: m%BMI 70–80% • Over 18: BMI 13–14.9	• Under 18: m%BMI >80%[36] • Over 18: BMI >15
HR (awake)	<40	40–50	>50
Cardio-vascular health[37 38]	Standing systolic BP below 0.4th centile for age or less than 90 if 18+, associated with recurrent syncope and postural drop in systolic BP of >20mmHg or increase in HR of over 30bpm (35bpm in <16 years)	Standing systolic BP <0.4th centile or <90 if 18+ associated with occasional syncope; postural drop in systolic BP of >15mmHg or increase in HR of up to 30bpm (35bpm in <16 years)	• Normal standing systolic BP for age and gender with reference to centile charts • Normal orthostatic cardiovascular changes • Normal heart rhythm

	Red: High impending risk to life	Amber: Alert to high concern for impending risk to life	Green: Low impending risk to life
Assessment of hydration status	• Fluid refusal • Severe dehydration (10%): reduced urine output, dry mouth, postural BP drop (see above), decreased skin turgor, sunken eyes, tachypnoea, tachycardia	• Severe fluid restriction • Moderate dehydration (5–10%): reduced urine output, dry mouth, postural BP drop (see above), normal skin turgor, some tachypnoea, some tachycardia, peripheral oedema	• Minimal fluid restriction • No more than mild dehydration (<5%): may have dry mouth or concerns about risk of dehydration with negative fluid balance
Temperature	<35.5°C tympanic or 35.0°C axillary	<36°C	>36°C
Muscular function[39]**: SUSS Test**	Unable to sit up from lying flat, or to get up from squat at all or only by using upper limbs to help (Score 0 or 1)	Unable to sit up or stand from squat without noticeable difficulty (Score 2)	Able to sit up from lying flat and stand from squat with no difficulty (Score 3)
Muscular function: Hand grip strength[40]	Male <30.5kg, Female <17.5kg (3rd percentile)	Male <38kg, Female <23kg (5th percentile)	Male >38kg, Female >23kg
Muscular function: MUAC[41]	<18cm (approx. BMI<13)	18–20cm (approx. BMI<15.5)	>20cm (approx. BMI >15.5)
Other clinical state	Life-threatening medical condition, e.g. severe haematemesis, acute confusion, severe cognitive slowing, diabetic ketoacidosis, upper gastrointestinal perforation, significant alcohol consumption	Non-life-threatening physical compromise, e.g. mild haematemesis, pressure sores	Evidence of physical compromise, e.g. poor cognitive flexibility, poor concentration

	Red: High impending risk to life	Amber: Alert to high concern for impending risk to life	Green: Low impending risk to life
ECG abnormalities	• **<18 years:** QTc >460ms (female), 450ms (male) • **18+ years:** QTc >450ms (females), 430ms (males) • And any other significant ECG abnormality	• **<18 years:** QTc >460ms (female), 450ms (male) • **18+ years:** QTc >450ms (females), >430ms (males). • And no other ECG anomaly • Taking medication known to prolong QTc interval	• **<18 years:** QTc <460ms (female), 450ms (male) • **18+ years:** QTc <450ms (females), <430ms (males)
Biochemical abnormalities[42]	• Hypophosphataemia and falling phosphate • Hypokalaemia (<2.5mmol/L) • Hypoalbuminaemia • Hypoglycaemia (<3mmol/L) • Hyponatraemia • Hypocalcaemia • Transaminases >3x normal range • Inpatients with diabetes mellitus: HbA1C >10% (86mmol/mol)		
Haematology	• Low white cell count • Haemoglobin <10g/L		
Disordered eating behaviours	Acute food refusal or estimated calorie intake <500kcal/day for 2+ days		

79. Question about 85 yrs old Alzheimer more than 65 ys old

   A. 2,
   B. 4,
   C. 8,
   D. 16
   E. 32 times.

   Risk of Alzheimer's dementia increases with age: 1% at age 60, 5% at age 65, doubles every 5 years, 40% of those aged 85.

80. Time limit for amphetamine testing.
   A. 24 hours,
   B. 3 days,
   C. 7 days

   Answer is 48 hrs

Substance	Time present in urine
Alcohol	Up to 12 hrs
Amphetamine	Up to 48 hours
Benzodiazepine	3 days (depending on t1/2)
Cannabis	Occasional use – up to 3 days. High daily use for long time – up to 4 weeks.
Cocaine	6 – 8 hrs; metabolites up to 2 - 4 days
Codeine	48 hours
Heroin	1 to 3 days
Methadone	3 days or more
Morphine	2 to 3 days
Phencyclidine (PCP)	3 to 8 days
LSD	Less than 24 hours (average users)

81. Prevalence of psychosis in diGeorge syndrome? 20%

   [SPMM]Mr David McDermott is a 40-year-old man with a diagnosis of DiGeorge syndrome. He has recently been referred to his local psychiatric clinic due to concerns about his mental health. Which of the following psychiatric disorders is most strongly associated with DiGeorge syndrome?

   Select one:

   A. Schizophrenia
   B. Depression
   C. Bipolar affective disorder
   D. Agoraphobia
   E. Body dysmorphic disorder

   Schizophrenia is most strongly associated, with up to 30% of those with DiGeorge syndrome developing a psychotic illness

82. Chi, Anova, ancova, mann whit, biserial something
   Phi, not point-biserial. It asked the correlation between two binary variables
   Kendall, Pearson, spearman, phi, point-biserial were options
   Pearson etc

83. What would reduce the risk of reoffending of indecent exposure?

   A. Antilibidinal,
   B. CBT,
   C. **Court appearance**,
   D. Expressing remorse,
   E. History of prior offences

84. Suicide within one year of self-harm attempt

A. 1 in 5,
B. 1 in 10,
C. **1 in 100**,
D. 1 in 500

Risk of suicide within one year of **DSH 0.7%** (nearly 1 in 100); more in males 1.1%; 0.5% in females. This is 66 x the general population risk. **Rise to 5% over 10 years**

85. Weight before and after? antipsychotics in same individuals. **This was Paired t test**
86. Cholesterol level in 3 groups before and after, One question was after adjusting for age, the other without.

87. ABI
    NNT
    NNH 7?
    ABI 10.2 or something?

88. LD cases:
    Fragile X
    Smith magenis
    Phenylketonuria
    Lesch nyhan (X-linked recessive)
    Down syndrome

Behavioural Phenotypes (2)
Choose one diagnosis for each of the behavioral phenotypes given below:

Shy, gaze avoidance and social anxiety	Choose...
Compulsively severely self-mutilating self-injurious behaviours such as biting of lips, inside of mouth and fingers	Choose...
Insatiable appetite, hyperphagia and obesity	Choose...
Stereotypies including self-hugging, severe self-injurious behaviours like head banging	Choose...
Oppositional and attention deficit problems in childhood	Choose...

Your answer is incorrect.

Explanation:

1) Fragile X syndrome is characterized by shyness, gaze avoidance, social anxiety (more characteristic), anxious inattentive, hyperactive, mimicking schizotypal disorder (males). 5-10% have an autistic disorder but most responsive to social cues and form attachments.

2) Lesch-Nyhan syndrome is associated with compulsive and severe mutilating self-injurious behaviour.

3) Around 80% of patients with Prader-Willi syndrome show hyperphagia. Affected individuals have an almost insatiable appetite leading to stealing of food and consume unpalatable food sometimes.

4) Individuals affected by Smith-Magenis syndrome have flattened midface, abnormally shaped upper lip, short hands and feet, abnormally placed ears often with recurring otitis media, protruding tongue, high arched palate, and moderate to severe mental retardation.

5) Oppositional defiance and attention deficit problems in childhood are behavioural phenotypes seen in some children with Downs syndrome

Fair skin, blue eyes, blond hair, mouse-like body odour, microcephaly, profound mental retardation

3) Phenylketonuria is due to a defect in phenylalanine hydroxylase (PAH) or cofactor (biopterin) with accumulation of phenylalanine transmitted in an autosomal recessive fashion. Symptoms are absent when the child is born but if undetected, a later development of seizures (25% generalized) and microcephaly with mild to profound mental retardation develops.

89. Emi about adding drugs for akathisia, EPS,.... propranolol,.....,...... : Akathisia, TD & Metabolic syndrome- if I remember correctly

Management of EPSEs

Choose the most likely treatment options for the following types of EPSEs

1. Pseudo-parkinsonism caused by trifluoperazine (choose 3 answers)

2. Acute dystonia with painful arching of the back caused by administration of halope

3. Akathisia induced by chlorpromazine (choose 3 answers)

- IM administration of benzodiazepines
- IM administration of anticholinergic drugs
- Reduce antipsychotic dose
- Increase antipsychotic dose
- Change to a typical antipsychotic drug
- Change to an atypical antipsychotic drug
- Consider clozapine
- Prescribe an oral anticholinergic
- Stop antidepressants if prescribed
- Propranalol 30-80 mg/day

Explanation:

Reducing antipsychotic dose, changing to an atypical antipsychotic drug and prescribing an oral anticholinergic medication are the options to manage drug induced parkinsonism. Anticholinergic drugs may be given orally, IM or IV depending on the severity of symptoms.

In extreme cases of acute dystonia the back may arch or the jaw may dislocate. IM administration of anticholinergic drugs is the preferred option in this situation.

For akathisia, several options including reducing antipsychotic dose, changing to an atypical antipsychotic drug and prescribing propranalol 30-80 mg/day or low dose clonazepam can be used. Anticholinergics are generally unhelpful.

Reducing antipsychotic dose and changing to an atypical antipsychotic drug are possible options for Tardive dyskinesia. Clozapine is the antipsychotic most likely to be associated with resolution of symptoms. Stop any anticholinergic drugs previously prescribed.

Correct Answer is : 1. Reduce antipsychotic dose, Change to an atypical antipsychotic drug, Prescribe an oral anticholinergic 2. IM administration of anticholinergic drugs 3. Reduce antipsychotic dose, Change to an atypical antipsychotic drug, Propranalol 30-80 mg/day 4. Reduce antipsychotic dose, Change to an atypical antipsychotic drug, Consider clozapine

90. Likelihood ratio for negative formula

[SPMM] The likelihood ratio for a negative test when sensitivity is 60%, and specificity is 80%.

Explanation: The likelihood ratio of negative test = **[(1-sensitivity)/specificity]** = [(1-60%)/80%] = [(1-0.6)/0.8] = 0.4/0.8 = 1/2 = 0.5

Q. [Psych-Mentor] A new blood test is developed for diagnosing bowel cancer and is compared to the gold standard of biopsy. The study reveals that the test has a sensitivity 60% of and a specificity of 80%. A patient receives a negative test result. What is the likelihood ratio for a negative test result?

A. 1.3 (1 dp)
B. 0.48
C. 0.5
D. -0.2
E. 0.3 (1 dp)

Explanation: LR- = (1 - sensitivity) / specificity
LR- = (1 - 0.6) / 0.8
LR- = 0.5
This means that the probability of having a negative test for individuals with the disease is 0.5 times or half of that of those without the disease.

91. Clozapine, sedation & motivation... strongest link*

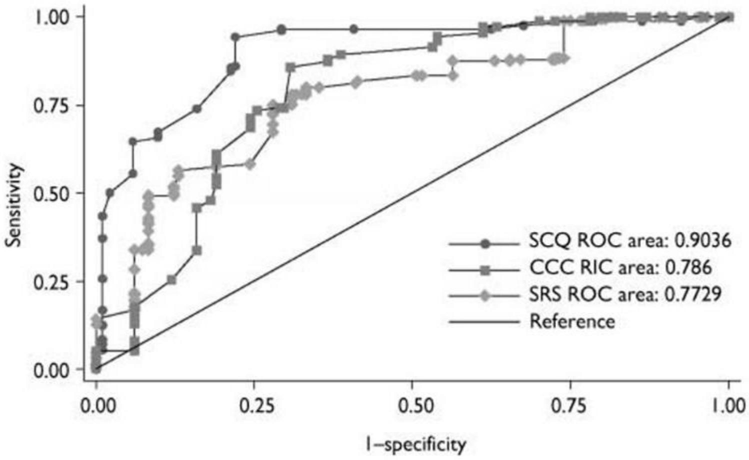
Very common side effect of clozapine is **sedation** or drowsiness. This occurs in most patients when they are new to clozapine as they titrate the dosage up. Sedation is not always a problem, since early in treatment with clozapine, people are often agitated or psychotic, and sedation can be calming

We found evidence that clozapine directly improves motivation, but this is opposed by the negative (but smaller) effect of clozapine-induced sedation on motivation.

92. ROC curve for SCQ, SRS and CCQ. Inference. Fresh q. SPMM
Scq has more specificity than srs and ccq [Check Q. 1 Mock 1 SPMM] psychmentor Q 387

Q. Which screening test offers the best compromise between the sensitivity and specificity at the cut-off points used in the study? **SCQ**

Q. Which of the screening instruments provides the best overall accuracy in discriminating between ASD and non-ASD? **SCQ**



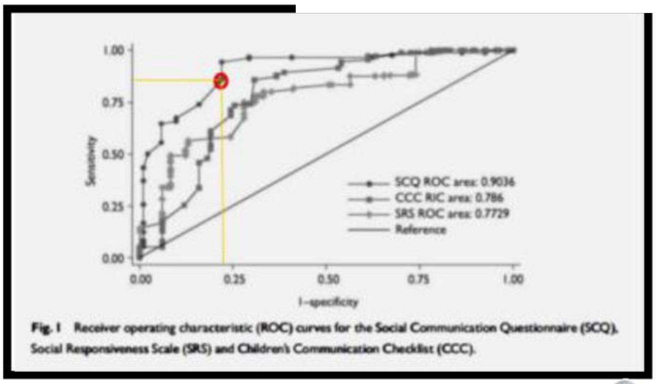


Fig. 1 Receiver operating characteristic (ROC) curves for the Social Communication Questionnaire (SCQ), Social Responsiveness Scale (SRS) and Children's Communication Checklist (CCC).

93. EMI- new diagnostic test- 3 questions- Sn, PPV, LR were the answers I think. **Definitions of these terms**.

Theme: Screening test statistics [Psych-Mentor]

A. Specificity
B. Relative risk
C. Absolute risk reduction
D. Sensitivity
E. Negative predictive value
F. Odds ratio
G. Likelihood ratio for a positive test result
H. Positive predictive value
I. Likelihood ratio for a negative test result
J. Relative risk reduction

Q. How much the odds of the disease increase when a test is positive

CORRECT ANSWER : **Likelihood ratio for a positive test result**

Q. Proportion of patients without the condition who have a negative test result

CORRECT ANSWER : **Specificity**

Q. The chance that the patient has the condition if the diagnostic test is positive

CORRECT ANSWER : **Positive predictive value**

94. Screening questionnaire which research has shown is good for adults. What would make you use it for the general population?

   A. **Sensitivity**.
   B. Accuracy,
   C. PPV,
   D. NPV, etc

95. Question said what would make it PD and not Drug induced?
symmetrical**, ?gets worse with time**, Then symmetrical won't be the answer

96. EMI- cleft lip/palate...what were the other questions in this set?

[SPMM] Tina Keyes is a 2-year-old girl who was born with a cleft palate, which is in the process of being surgically repaired. She demonstrated evidence of development delay during her first year and it is now thought that she will likely have a degree of intellectual disability. Which of the following medications taken during pregnancy is most likely to have contributed to these problems?

Select one:

   A. Diazepam
   B. Lithium
   C. Sertraline
   D. Sodium valproate
   E. Clozapine

Drugs in perinatal psychiatry
Which of the drugs provided in the options menu, when taken by a mother while pregnant, is most likely to be implicated in each case scenario?

1. 1 year old with cleft palate and delayed milestones ✖

2. A child with microcephaly, challenging behaviour, and cognitive deficits who weighed 2kg at birth ✖

3. Jittery newborn with overarousal and tachypnoea 2 hours after delivery ✖

Explanation: There is strong evidence that valproate is associated with increased risk of congenital and major congenital malformations (as many as 11% of exposed children); cleft lip and/or palate, neural tube defects (risk of spina bifida 1-2%), congenital heart anomalies, radial ray defects, ophthalmological and genitourinary anomalies; developmental delay (delayed walking and talking, memory problems, difficulty with speech and language, lower intellectual ability), and autism.

Clinical features of fetal alcohol spectrum disorder include microcephaly, abnormal facial features, growth deficits, mild intellectual disability, and associated behavioural problems (such as hyperactivity and sleep problems).

Stimulant use (such as cocaine) prior to delivery is associated with a neonatal withdrawal syndrome including symptoms of agitation, vomiting, and tachypnoea.

Jittery, crying a lot – opioid
Tachypnoea and tachycardia – Cocaine
Increase need for Handling – Nicotine
Table page 15 (Q. 33)

October 2023

97. Question was to calculate OR for daily use for FEP. There was a stats question of cannabis skunk like , what was the odds ratio in that?

An environmental agency identifies 200 patients with life time history of panic attacks and 200 matched controls without the condition. Exposure to loud traffic noises was compared in two groups. 20 patients and ten controls had such an exposure. In the above study the odds ratio of being exposed to loud noise in those who have panic attacks compared to those who do not is

Select one:
- (20*180)/(10*190)
- (10*190)/(20*180)
- (20*190)/(10*180)
- (20*10)/(400)
- (20*200)/(10*200)

Your answer is incorrect.

To calculate odds ratio, one must construct a 2X2 table. See the attached image.

	Panic Disorder	No panic disorder	total
Loud noise exposed	20	10	30
Not exposed	180	190	370
total	200	200	400

Case exposure rate : 20/200

Control exposure rate : 10/200

From this table, the cross product (odds ratio) will be (20*190)/(10*180)

The correct answer is: (20*190)/(10*180)

98. Most significant predictor of FEP -frequency of use, other options not statistically significant. **Q21 mock 1**

	First-episode psychosis group (n=410)	Control group (n=370)	p value
Total population			
Lifetime history of cannabis use	..	..	0·277
Yes	275 (67%)	232 (63%)	..
No (never used)	135 (33%)	138 (37%)	..
Frequency of use	..	..	<0·0001
Less than once per week	68 (17%)	128 (35%)	..
At weekends	84 (20%)	63 (17%)	..
Every day	123 (30%)	41 (11%)	..
Most used type of cannabis	..	..	<0·0001
Never used	135 (33%)	138 (37%)	..
Hash-like	57 (14%)	162 (44%)	..
Skunk-like	218 (53%)	70 (19%)	..
Cannabis users			
Duration of use (years)	9·7 (7·4)	9·1 (7·8)	0·635
No details	3	1	..
Age at first cannabis use (years)	16·1 (4·2)	16·6 (3·2)	0·146
No details	3	1	..
Age at first use ≤15 years	..	..	0·028
No	172 (63%)	178 (77%)	..
Yes	100 (36%)	53 (23%)	..
No details	3	1	..

Data are n (%) or mean (SD) unless stated otherwise.

***Table 2:* Cannabis use**

99. Elderly lady with 6th Nerve palsy and Nystagmus, in and out of consciousness...was it **subdural hematoma**? Wernicke's encephalopathy

100. Covid and mental illness outcomes table, after adjusting for demographics and age – only schizophrenia was significant. **Searching for the study…..**

101. Multiple imputation definition

[SPMM] **Multiple imputation**: fills in missing values by generating plausible numbers derived from distributions of and relationships among observed variables in the data set.

[Google] There are many ways to approach missing data. common approach is **imputation**. Imputation simply means replacing the missing values with an estimate, then analysing the full data set as if the imputed values were actual observed values. The following are common methods:

Mean imputation

Simply calculate the mean of the observed values for that variable for all individuals who are non-missing. It has the advantage of keeping the same mean and the same sample size, but many, many disadvantages. Pretty much every method listed below is better than mean imputation.

Substitution

Impute the value from a new individual who was not selected to be in the sample.

In other words, go find a new subject and use their value instead.

Hot deck imputation

A randomly chosen value from an individual in the sample who has similar values on other variables.

In other words, find all the sample subjects who are similar on other variables, then randomly choose one of their values on the missing variable. One advantage is you are constrained to only possible values. In other words, if Age in your study is restricted to being between 5 and 10, you will always get a value between 5 and 10 this way. Another is the random component, which adds in some variability. This is important for accurate standard errors.

Cold deck imputation

A systematically chosen value from an individual who has similar values on other variables. This is similar to Hot Deck in most ways, but removes the random variation. So for example, you may always choose the third individual in the same experimental condition and block.

Regression imputation

The predicted value obtained by regressing the missing variable on other variables. So instead of just taking the mean, you're taking the predicted value, based on other variables. This preserves relationships among variables involved in the imputation model, but not variability around predicted values.

102. **Heroin.** Read somewhere it's most common cause of substance related **jittery** baby

> ☞ **Jittery ➔ think Opioids or Cocaine**
> **If overarousal, tachycardia and tachypnoea ➔ Cocaine withdrawal**
> **High-pitching crying, irritability, tremors ➔ Opioids withdrawal**

103. Dehydration. I think hr cut off is 40 in MEED. **Look Q. 78**
104. Patient on clozapine for 3 months, still having auditory hallucinations, next best step.

A. **Check adherence**,
B. Increase dose,
C. Augment with amisulpride, etc

In situations where clozapine has been ineffective (and all other conventional options have been tried) the following options are available (see Maudsley for complete list):-

- Allopurinol + antipsychotic
- Max dose Amisulpride (1200mg/day)
- Max dose aripiprazole 30 mg/day
- D-Alanine and D-Serine
- ECT
- High dose olanzapine
- Olanzapine with various combinations

105. Differentiating bulimia from anorexia: **Poor impulse control**, laxative abuse, etc

[SPMM] Which of the following suggests a diagnosis of bulimia nervosa rather than anorexia nervosa? Select one:

A. Age of onset at 15
B. Lanugo hairs
C. Calorie counting
D. Laxative abuse
E. Poor impulse control

Explanation: There is an increased prevalence of disorders of impulsivity and lack of control in people with bulimia nervosa. Patients who have to satisfy their binges at all costs (e.g. shoplifting) may have issues with impulsivity and rage. People with bulimic illnesses also have a higher prevalence of substance misuse

106. EMIs- urinary frequency and urgency, which substance – **Ketamine**

[SPMM] A man who has been abusing multiple substances now complains of frequent urination and passing blood in the urine. The most likely offending drug is

Select one:

A. Cannabis
B. Heroin
C. Crack
D. LSD
E. Ketamine

Explanation: Ketamine is structurally similar to phencyclidine and is used as an anaesthetic. It is a unique anaesthetic, as it does not depress the reticular activating system (thus does not usually knock out consciousness completely) but reduces the cortical awareness of painful stimuli. It is taken illicitly as a sniffed powder. The use of ketamine can cause urinary tract symptoms (e.g**. frequency, urgency, urge incontinence, dysuria, and haematuria**). The causal agent has not been determined, but direct irritation by ketamine and/or its metabolites is a possibility. Investigations have revealed interstitial cystitis, detrusor overactivity, and decreased bladder capacity. Symptoms generally settle several weeks after stopping ketamine. **Up to 30% of Ketamine users have urinary symptoms.**

107. What type of cost for a patient who could not go to work when ill. **Indirect cost**

[SPMM] Due to the difficulty in recruiting a full-time nurse for the ECT clinic, your hospital decides to provide the nurse who works at pain clinic to attend the ECT service on 'secondment '. Select the term that best describes the costs incurred by patients due to the absence of a nurse at the pain clinic **as a result of this arrangement.**

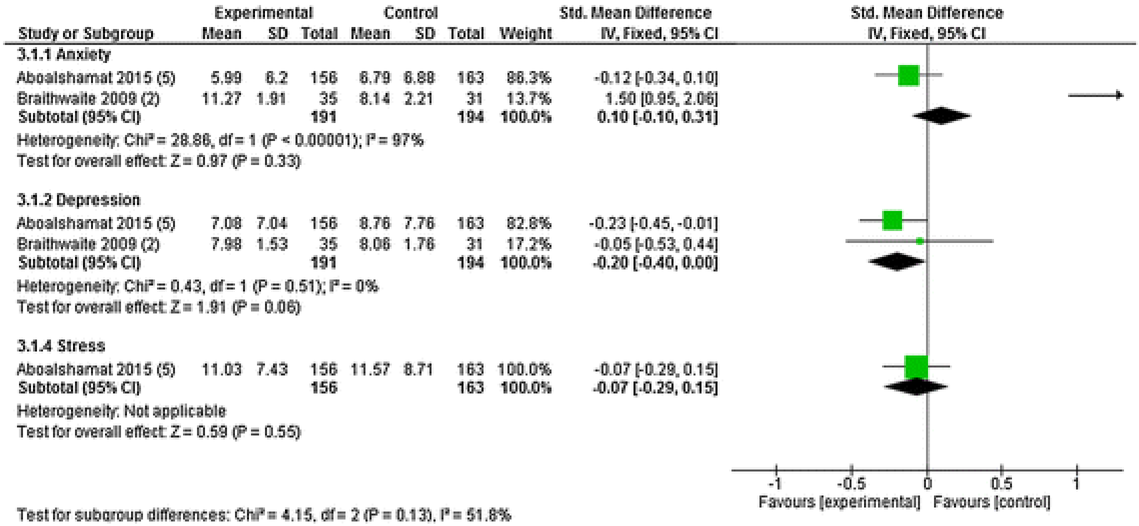
Select one:

A. Indirect cost
B. Direct cost
C. Zero-sum cost
D. Intangible cost
E. Opportunity cost

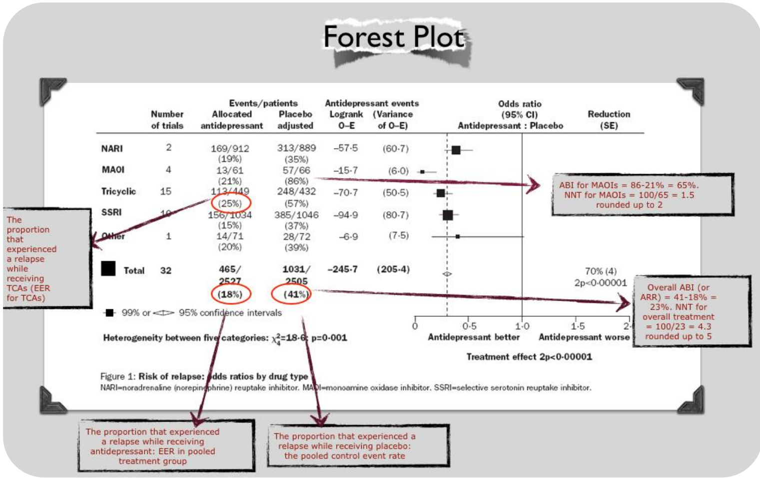
The opportunity cost is defined as the benefits that must be foregone when making any **resource allocation decision.** It refers to the fact that expending resources on one health care issue inevitably reduces the investment that could be made elsewhere. The opportunity cost of employing an ECT nurse is the benefit we forego by being unable to support the pain clinic**. If the question asks about the cost borne by the patient then it will be Intangible.**

108. Stats- Forest plot question about some Psychoeducation - 3 questions were there-

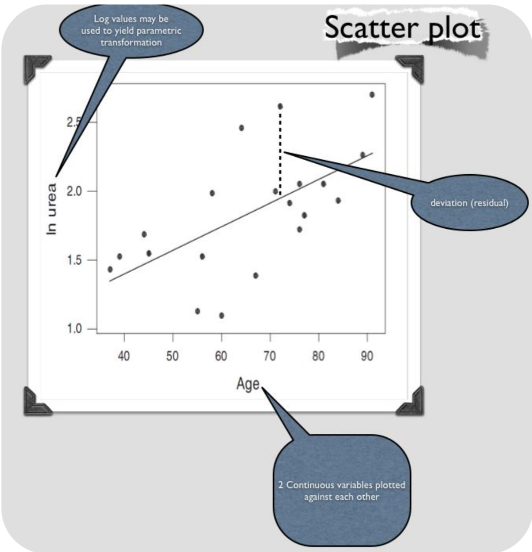
1. which study is significant?
2. Why is there an arrowhead in one study
3. ? Overall NNT



Forest plot of psychoeducational interventions compared to alternative education. Note the number in parentheses after the author and year denotes the quality assessment score out of a maximum of 10
https://www.researchgate.net/figure/Forest-plot-of-psychoeducational-interventions-compared-to-alternative-education-Note_fig2_315178868



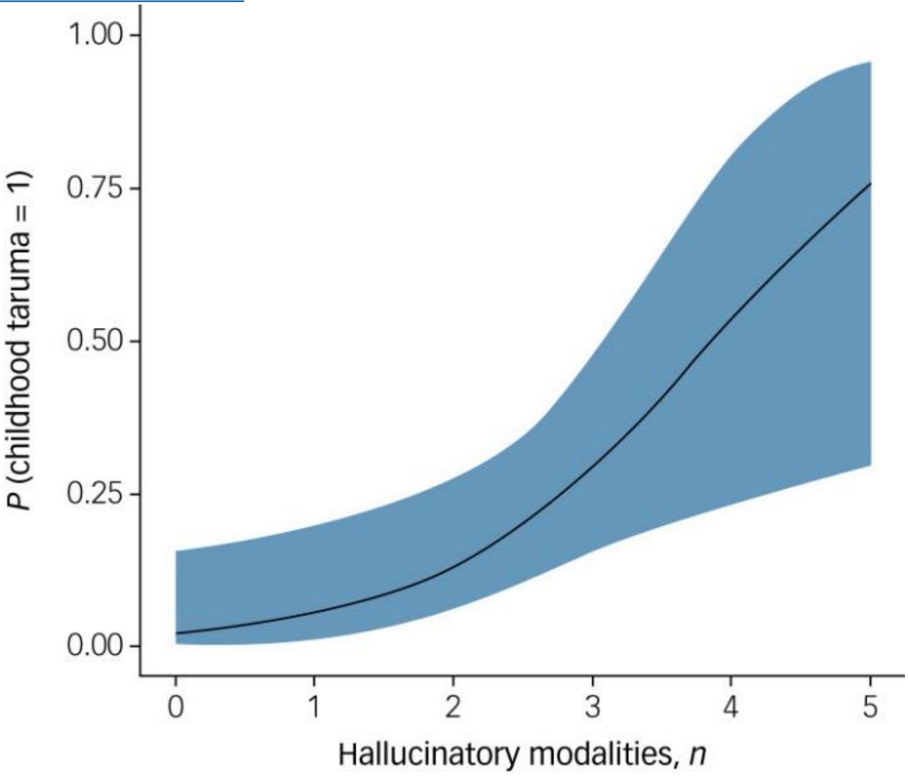
109. Stats- scatter plot with regression 0.9
Last one from the options



110. Stats- childhood trauma severity/number & modalities of hallucinations (blue coloured curve)
Severity of trauma to the hallucinations?

A. No. of trauma vs no. of modalities of Hallu
B. Severity of trauma vs no. Of modalities of Hallu
C. No of hallucinatory modalities Vs probability of trauma
D. Probability of hallucinations Vs no of trauma

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7557870/#:~:text=Estimated%20marginal%20means%20with%2095,0.017%2C%20n%20%3D%2075



Estimated marginal means with 95% confidence interval showing the positive relationship between **the number of sensory modalities in hallucinatory experiences and the probability of a childhood trauma** (OR = 2.24 (95% CI 1.16–4.33), $P = 0.017$, $n = 75$)

111. Stats - hazard ratio- dementia types (gray, yellow, pink) & age, some other factor adjusted stuff Most significant they were asking. Grey & Yellow ones weren't touching the line corresponding to the Hazard ratio of 1
In above 65 is that not vascular and Alzheimer's. Both don't cross line of no effect ?
But the question was statistically significant not association. both did not touch the line of no effect so should be both. Question mentioned statistical significance.

Q. Which disease got the highest Hazards ratio ?

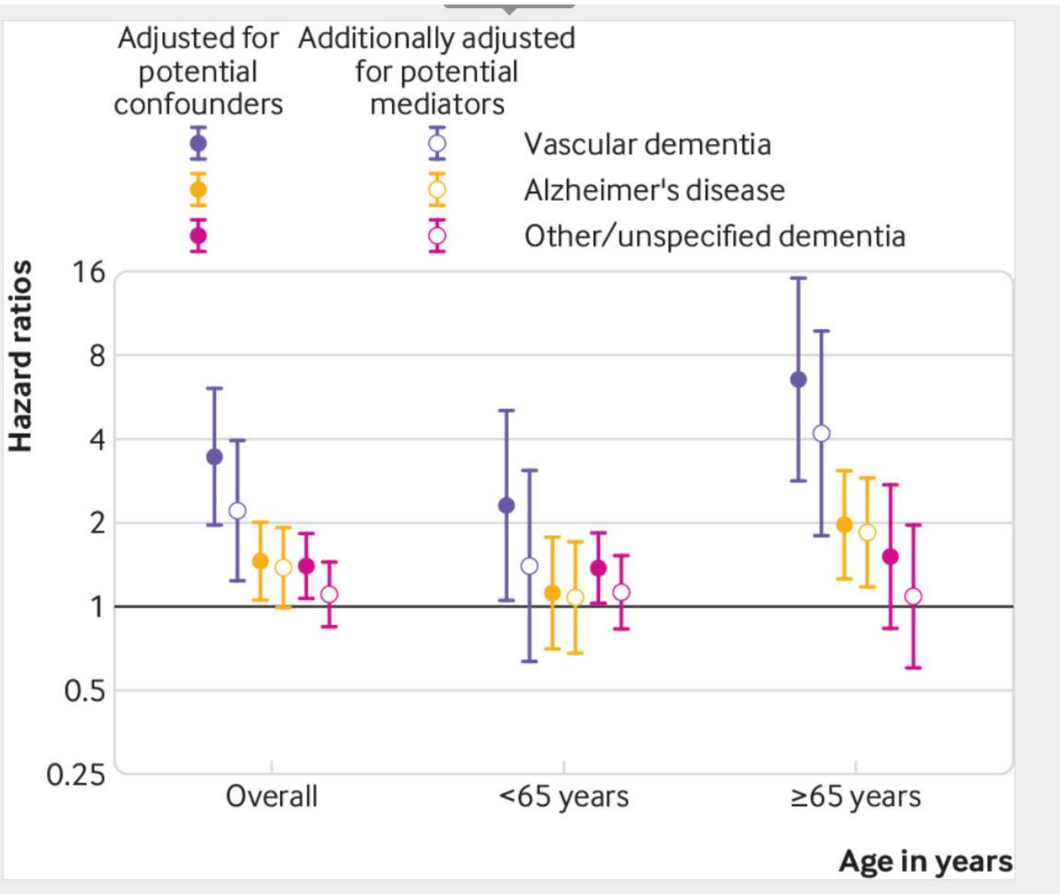


Fig 2
Associations between history of pre-eclampsia and dementia, overall, by dementia subtype, and by attained age, in cohort of women with ≥1 live birth or stillbirth in 1978-2015 in Denmark. Hazard ratios with 95% CIs compare risks of vascular dementia, Alzheimer's disease, and other/unspecified dementia in women with history of pre-eclampsia and women with no history of pre-eclampsia, overall and separately for women <65 and ≥65 years of age. All hazard ratios are adjusted for birth year (5 year intervals), parity (1, 2, ≥3 live births and/or stillbirths), and region of most recent delivery (potential confounders); age was underlying time scale in Cox model. Potential mediators additionally adjusted for were cardiovascular disease, stroke, hypertension, chronic kidney disease, and diabetes

https://www.bmj.com/content/363/bmj.k4109
From the study: Our finding that pre-eclampsia was strongly associated with vascular dementia adds convincing epidemiological evidence to suggest that the two conditions may share underlying mechanisms or susceptibility pathways. In contrast, we found little evidence to support an association between pre-eclampsia and Alzheimer's disease, particularly after accounting for the potential effect of obesity.

Answer: **Vascular Dementia**

112. Roman Catholics 55-74 age. Lowest suicides if I'm recalling correctly.
Denominations of Roman Catholic, Conservative Christians, Orthodox and no denomination with suicides grouped by ages (can't remember the ages exactly) with their odds.**: Suicide was higher in any group.**
Denominations of Roman Catholic, Conservative Christians, Orthodox and no denomination with suicides grouped by ages (can't remember the ages exactly) with their odds. **: Suicide was higher in any group** [we think stat question]

The question could be: Which group has highest suicide rate and which has lowest suicide rate. We are not sure whether they mention an age group or not.

	Age 16–34 years	Age 35–54 years	Ag 55–74 y
Suicides in group, *n*	423	518	178
Suicide rate (95% CI)			
Roman Catholic	15.1 (13.2–17.2)	16.7 (14.6–19.0)	7.4 (5.7–9.7)
Protestant	10.1 (8.5–12.1)	14.8 (12.9–17.0)	8.0 (6.5– 9.8)
Conservative Christian	9.2 (5.8–14.6)	6.5 (3.9–11.1)	8.2 (4.6–14.4)
No religion	14.4 (11.3–18.3)	17.3 (13.9–21.5)	11.8 (7.8–17.7)

Table 3 Age-specific suicide rates per 100 000 population categorised by religious affiliation

Table 4 shows the hazard ratios from fully adjusted Cox proportional hazards models stratified by age group. These mirror

a. Hazard ratios from separate models fully adjusted for all covariates in Table DS1.

	Hazard ratio (95% CI)[a]		
Affiliation	Age 16–34 years	Age 35–54 years	Ag 55–74 years
Roman Catholic	1.00	1.00	1.00
Protestant	0.73 (0.58–0.92)	1.04 (0.86–1.26)	1.22 (0.87–1.73)
Conservative Christian	0.75 (0.46–1.21)	0.50 (0.29–0.85)	1.31 (0.90–2.46)
No religion	0.88 (0.67–1.17)	1.01 (0.78–1.32)	1.41 (0.86–2.32)

Table 4 Risk of suicide according to denominational affiliation stratified according to age

Age	Before stratification		After Stratification	
	High	**Low**	**High**	**Low**
16 – 34	**Roman**	Christian	**Roman**	Protestant lower with a very low margin than Christian 0.02
35 – 54	No religion	Christian	All groups almost same rate exp for Christian	Christian Significantly low
55 – 74	No Religion (11.8)	Roman	No Religion	Roman

113. Denominations of Roman Catholic, Conservative Christians, Orthodox and no denomination with suicides grouped by ages (can't remember the ages exactly) with their odds. He asks about higher suicide or lower suicide.
Roman Catholics 55-74 age. Lowest suicides if I'm recalling.

114. Power of study for sample size collection. Was there a question along this line?
Stats- what explains **Cost benefit study best**?- outcomes in 'monetary terms'

Q. Which of the following is true regarding cost-benefit analysis? [Psych-Mentor]

A. It involves the use of the DALY
B. Provides a measure given monetary terms
C. It involves the use of the QALY
D. It not appropriate for use in psychiatric studies
E. Relates costs to a single clinical measure of effectiveness

There are four basic methods of economic evaluation that you need to be familiar with:

- Cost-effectiveness analysis (CEA)
- Cost-benefit analysis (CBA)
- Cost-utility analysis (CUA)
- Cost-minimisation analysis (CMA)

All four are identical in their approach to capturing costs, however, they differ in how they assess health effects.

Cost-effectiveness analysis (CEA)

CEA compares a number of interventions by relating costs to a single clinical measure of effectiveness (e.g. symptom reduction, improvement in activities of daily living).

cost-effectiveness ratio = total cost / units of effectiveness

For example consider this study on anorexia which compared the costs of three approaches to treatment (inpatient, specialist outpatient, and general outpatient) per improvement on the Morgan Russell Average Outcome Scale (Byford, 2007).

Other examples:

- £'s spent per lives saved
- £'s spent per each depression free day

CEA is generally done when CBA cannot be performed due to the inability to monetise benefits.

Combining both costs and effects, the findings of a CEA are usually reported as an **incremental cost-effectiveness ratio (ICER).**

ICER intervention A versus intervention B = (costs A - costs B) / (effects A - effects B)

A treatment that is found to be BOTH more costly AND less effective is said to be 'dominated' by the other treatment.

The advantage of expressing health outcomes in natural units is that these are often observable, relatively easy to measure and, often, meaningful to clinicians. The disadvantage, however, is that they limit the scope of comparisons. For example, cost-effectiveness analyses using outcomes such as depression-free days achieved only allow comparisons with other interventions that can be expressed using exactly the same metric. Survival-related outcomes such as life years gained allow comparisons over a broader range of conditions, but they disregard morbidity and quality of life.

Cost-benefit analysis (CBA)

CBA is a technique in which all the costs and benefits of an intervention are measured in terms of money. A CBA is used to establish which of the alternatives has the greatest net benefit.

CBA requires that all the consequences of an intervention, such as life-years saved, treatment side-effects, symptom relief, disability, pain and discomfort, are allocated a monetary value.

CBA is relatively rarely used in mental health service evaluation mainly due to the difficulty in converting benefits from mental health programmes into monetary values.

Cost-utility analysis (CUA)

CUA is a special form of CEA in which health benefits / outcomes are measured in broader, more generic ways enabling comparisons between treatments for different diseases and conditions.

Multidimensional health outcomes are measured by a single preference- or utility-based index such as the QALY.

Quality-Adjusted-Life-Years (QALYs). QALYs are a composite measure of gains in life expectancy and health-related quality of life. One QALY is equal to 1 year of life in perfect health. QALYs are calculated by estimating the years of life remaining for a patient following a particular treatment or intervention and weighting each year with a quality-of-life score (on a 0 to 1 scale). The quality of life score can be negative in situations where the outcome is considered to be worse than death (death = 0 on the scale).

CUA offers something that CEA cannot, which is to compare across treatments for different conditions. In principle, it is possible to compare treatments for, say, cancer with, say, schizophrenia to determine which is the most efficient at producing health gain in the form of QALYs.

Findings of CUA are often reported as an incremental cost-utility ratio (ICUR).

ICUR intervention A versus intervention B = (costs A - costs B) / (QALYs gained by A - QALYs gained by B)

Cost-minimisation analysis (CMA)

An economic evaluation in which consequences of competing interventions are the same and in which only inputs, that is, costs are taken into consideration. **The aim is to decide the least costly way of achieving the same outcome.**

Costs

There are three main types of costs in economic evaluation studies:

- Direct - those associated directly with the healthcare intervention (e.g. staff time, medical supplies, cost of travel for the patient, childcare costs for the patient, costs falling on other social sectors such as domestic help from social services)
- Indirect - those incurred by the reduced productivity of the patient (e.g. time of work, reduced work productivity, time spent caring for the patient by relatives)
- Intangible - those that are difficult to measure (e.g. pain or suffering on the part of the patient)

115. Link between - Clozapine dose, sedation & Motivation...? **(mock 14 Q. 13)**

116. : Most common disorder seen in patients attending psychiatric rehabilitation services?
Substance misuse disorder **other options** might be **Schizophrenia**, mood disorder, personality and neurodevelopmental.

A national survey of inpatient mental health rehabilitation services across England found that 80% of those using these services had a diagnosis of a psychotic illness, usually schizophrenia or schizoaffective disorder.

In the UK, what is the most common type of mental disorder in patients on mental health rehabilitation units?
Select one:

A. Intellectual disability
B. OCD
C. Psychotic disorder
D. Anxiety disorder
E. Mood disorder

Explanation : The most common disorders in patients on mental health rehabilitation disorders are psychotic disorders, including **schizophrenia and schizoaffective disorder**. Bipolar disorder is also relatively common.

117. ICD11 PD vignette- man who was fired multiple times, feels he has enough skills, starts talking about the trip he booked in the morning - answer? **Dissociable** or **disinhibited**? **New ICD11.**

ICD-11 trait domain	Description
Negative Affectivity	The core feature of the Negative Affectivity trait domain is the tendency to experience a broad range of negative emotions. Common manifestations of Negative Affectivity, not all of which may be present in a given individual at a given time, include: **experiencing a broad range of negative emotions with a frequency and intensity out of proportion to the situation** **(borderline);** emotional lability and poor emotion regulation; negativistic attitudes; low self-esteem and self-confidence; and **mistrustfulness.** **(paranoid)**
Detachment	The core feature of the Detachment trait domain is the tendency to maintain interpersonal distance (social detachment) and emotional distance (emotional detachment). Common manifestations of Detachment, not all of which may be present in a given individual at a given time, include: social detachment (**avoidance of social interactions**, **lack of friendships, and avoidance of intimacy**); and emotional detachment (reserve, aloofness, and limited emotional expression and experience). **(schizoid)**
Dissociality	The core feature of the Dissociality trait domain is disregard for the rights and feelings of others, encompassing both self-centeredness and lack of empathy. Common manifestations of Dissociality, not all of which may be present in a given individual at a given time, include: self-centeredness (e.g., **sense of entitlement**, expectation of others' admiration, positive or negative attention-seeking behaviours, concern with one's own needs, desires and comfort and not those of others); and **lack of empathy** (i.e., indifference to whether one's actions inconvenience hurt others, which may include being deceptive, manipulative, and exploitative of others, being mean and physically aggressive, callousness in response to others' suffering, and ruthlessness in obtaining one's goals).
Disinhibition	The core feature of the Disinhibition trait domain is the tendency to act rashly based on immediate external or internal stimuli (i.e., sensations, emotions, thoughts), without consideration of potential negative consequences. Common manifestations of Disinhibition, not all of which may be present in a given individual at a given time, include**: impulsivity; distractibility; irresponsibility; recklessness; and lack of planning.** **(impulsive type BPD)**

ICD-11 trait domain	Description
Anankastia	The core feature of the Anankastia trait domain is a narrow focus on one's rigid standard of perfection and of right and wrong, and on controlling one's own and others' behaviour and controlling situations to ensure conformity to these standards. Common manifestations of Anankastia, not all of which may be present in a given individual at a given time, include: perfectionism (e.g., concern with social rules, obligations, and norms of right and wrong, scrupulous attention to detail, rigid, systematic, day-to-day routines, hyper-scheduling and planfulness, emphasis on organisation, orderliness, and neatness); and emotional and behavioural constraint (e.g., rigid control over emotional expression, stubbornness and inflexibility, risk-avoidance, perseveration, and deliberativeness). **OCD**

118. Mirtazapine adverse effect study- patients left the study when they came to know about the side effect - how to prevent this bias? **blinding**

[SPMM]**Measurement Bias** results when data are not collected in a uniform fashion. For e.g. when cases are interviewed in person while controls are interviewed over the telephone. This type of bias is minimised by "**blinding**".

- **Selection** Bias: Allocation Concealment, Stratification, Minimisation, Block and cluster randomization
- **Measurement** bias – Blinding

Terms used in RCT
Identify the terms commonly used for each of the descriptions given below.

This refers to maintaining secrecy with regard to the allocation at the start of the study.

This term refers to maintaining secrecy about the status of a patient as to which arm he or she belongs, once the allocation is complete.

Allocation of each participant is done methodically to ensure balance of confounding factors between groups

This process may distribute even unknown confounders equally across both groups.

Choose...
Matching
Intention to treat analysis
Minimization
Confounder adjustment
Randomization
Allocation concealment
Surrogate end points
Blinding
Stratification

1) Allocation concealment refers to maintaining secrecy with regard to the allocation itself at the start of the study.
2) Blinding refers to maintaining secrecy about the status of a patient as to which arm he or she belongs, once the allocation is complete.
3) Minimization is the allocation of each participant aims to ensure the balance of prognostic or confounding factors between groups.
4) Randomization may distribute both known and unknown confounders equally across both groups and thus reduce any confounding bias.

In an allocation scheme, the next allocation of a subject depends on the characteristics of those already allocated. This scheme can be called
Select one:

A. Haphazard randomisation scheme
B. Stratification scheme
C. Blocking scheme
D. Minimisation scheme
E. Matching scheme

The minimization method was described by Taves and Pocock and Simon. Taves' article coined the term "minimization." Important prognostic factors are identified before the trial starts, and assignment of a new patient to a treatment group is determined so as to minimize the differences between the groups in terms of these factors. Consider a hypothetical trial with 16 patients already recruited. If the 17th patient considered is male, aged 38 and with a high risk factor, the decision to allocate to the treatment or control group can be made by comparing the total burden of distribution of the confounders for each category and choosing the group that gives most balance overall.

119. 3 back to back questions on Bias-
ITT controls which kind of bias- **attrition bias**
What were the other 2 questions here? it asked what analysis is important to look at when there's a huge difference in the numbers of participants discontinued in both groups, unsure if it was best case analysis or intention to treat.

[SPMM]100 patients are enrolled in an RCT comparing an herbal medicine with fluoxetine at 12 weeks for depression. Out of 50 who were assigned to take fluoxetine, 8 did not turn up for follow-up. Out of 50 who were assigned to herbal treatment, 12 did not complete the treatment or missing. If the above study reports numbers improved out of all 100 randomised, which of the following processes has been carried out?
Select one:

A. Per protocol analysis
B. Multiple regression analysis
C. Stratification analysis
D. Intention to treat analysis
E. Treatment received analysis

Explanation: This is called intention to treat analysis. The rationale is that in most RCTs, one sets out to estimate the broader effects of allocating an intervention in clinical practice, rather than just the narrow effects in the subgroup of participants who adhere to it. The ITT strategy gives a conservative estimate of the treatment effect compared with what would be expected in an ideal world of full compliance. Hence, ITT tends to make RCT results more generalizable and pragmatic.

Intention to treat
Identify the methods of intention to treat in an RCT from the descriptions given below

Assume drop-outs in active arm have negative outcomes and drop-outs in control arm have positive outcomes

Assume there was no change in the available ratings after drop-out

Assume that the scores for the drop-out subjects were same as the average for their groups

Choose...
Differential attrition
Worst case scenario
Last observation carried forward
Survival curve
Concealment allocation
Multiple imputations
Minimization
Mean imputations
Stratification

Your answer is incorrect.

Explanation: The methods of intention to treat include 1) **worst-case scenario**: Assume drop-outs in active arm have negative outcomes and dropouts in control arm have positive ones 2) **Last observation carried forward approach** assumes that the last score in a rating scale available from a subject who dropped out continued till the end of the trial and had no further change. 3) **Mean imputations-** Assume that the score for a subject who dropped out is same as the average score in a rating scale available from the rest of the group (Ref: Oxford handbook of psychiatry- pg 110)

The correct answer is: Assume drop-outs in active arm have negative outcomes and drop-outs in control arm have positive outcomes → Worst case scenario, Assume there was no change in the available ratings after drop-out → Last observation carried forward, Assume that the scores for the drop-out subjects were same as the average for their groups → Mean imputations

120. What's the best way to have ITT?

A. Worst case scenario,
B. Imputation,
C. Per protocol analysis

Alternatives to ITT: [SPMM]

Per-protocol (PP) analysis: According to this **only patient who sufficiently complied with the trial's** protocol are considered in the final analysis; the antithesis of intention to treat analysis. It is also called "on treatment" analysis. PP analysis is often carried out in biological explanatory RCTs and as the secondary analysis in effectiveness trials.

Treatment-received (TR) analysis: Another approach is to analyse subjects according to the actual treatment received – not actual allocation made. But the effect of a random assignment is compromised here due to contamination of trial by one group of subjects moving to the other group.

ITT analysis **must be the norm** unless there is overwhelming justification for a different analysis policy (e.g. high proportion of participants found not to have the disease etc.)

121. Power is used for

A. Type 1 error
B. P value
C. **Sample size**

[SPMM] A new drug is being tested to treat akathisia induced by neuroleptics. All of the following measures will improve the statistical power of this RCT except

Select one:

A. Reducing drop-out rates by monitoring adherence
B. Selecting more homogenous clinical sample
C. Enrolling more patients in the study
D. Requiring a significantly high difference in proportion of responders between placebo and active arms
E. Conducting subgroup analysis on patients with a shorter duration of illness

Explanation: The power of a study to detect a difference that exists in the population increases with increasing sample size. It reduces with increasing variance in the sample. Power is (1 - ß); it reduces as the significance level is made smaller (i.e., alpha or p is set too low) provided other factors are kept constant. Power increases as the absolute value of the distance between the null and alternative means increases (effect size).

[SPMM] Which of the following measures increases the power of a study?

Select one:

A. Including heterogenous patient groups
B. Using more than 2 primary outcome measures
C. Comparing two active treatments
D. Longer period of follow-up in a trial
E. Comparing active treatment vs. placebo

If active treatment and placebo are compared, the effect may be bigger resulting in a powerful study

Q. Which one of the following statements regarding the power of a study is correct? [Psych-Mentor]

A. Is the probability of rejecting the null hypothesis when it is false.
B. Decreases with increasing sample size
C. Lies within 2 standard deviations of the mean
D. Is the chance a significant p value will be reached Is equal to 1 - (the probability of a type I error)

Stats Power **The power of a study is the probability of (correctly) rejecting the null hypothesis when it is false** (i.e. it will not make a Type II error). Basically we use the power to help us decide how many people need to be recruited in a study in order to detect a clinically meaningful difference or effect. **Power can assume values between 0 and 1** (Since probability values are expressed by numbers between 0 and 1 only). Sometimes it is expressed as a percentage - 0 referring to 0 %, and 1 referring to 100 %. Power is expressed as **1 - beta,** where beta is the probability of a Type II error. A power 0.80 is often seen as the level of minimum acceptability. Power is influenced by the following:- Sample size (larger samples lead to parameter estimations of smaller variance and therefore increase the study's ability to detect a significant effect Meaningful effects size (this has to be decided at the beginning of a study, it is the size of the difference between two means that lead you to reject the null hypothesis) Significance level (aka the alpha level, which is the probability of a type I error)

Error Type	Description	Symbol
Type 1 Error *False Positive*	Incorrectly rejecting a null hypothesis when it's true.	α (Alpha)
Type 2 Error *False Negative*	Incorrectly failing to reject a null hypothesis when it's false.	β (Beta)

Q. Which of the following is true regarding statistical power?

A. The power of a study is equivalent to the type I error
B. Power is a measure of a studies ability to change clinical practice
C. Power can assume any value from -1 to 1
D. The larger the sample size of a study the greater the power
E. The term power is synonymous with the term significance

122. There was a study, question was asking about what is true after adjusting for confounders. Which had the highest effect after adjusting for confounders?

123. EMI on Drugs & Birth anomalies...Cleft lip, Developmental delay –
    A. **Antenatal valproate exposure**.

124. Microcephaly, born 2 kg - **antenatal alcohol exposure**

125. Patient with inability to verbalize with some weird movement when touching upper lip, which part of the brain affected? Options were

    A. Frontal lobe,
    B. Temporal, parietal,
    C. Occipital, and
    D. Cerebellum

Frontal release sign. Snout reflex

The **suck reflex** is elicited by lightly touching or tapping on the lips with an object such as a tongue blade, reflex hammer, or the examiner's finger. At times the reflex is obtained merely by approaching the lips with an object.

The **snout reflex** is brought about by tapping the upper lip lightly. The contraction of the muscles causes the mouth to resemble a snout. **Also seen in dementia and prolonged antipsychotics (Phenothiazines)**

The **palm omental reflex** occurs when a disagreeable stimulus is drawn from the thenar eminence at the wrist up to the base of the thumb. There is ipsilateral contraction of the orbicularis oris and mentalis muscles. The skin over the chin wrinkles, and the corner of the mouth elevates slightly.

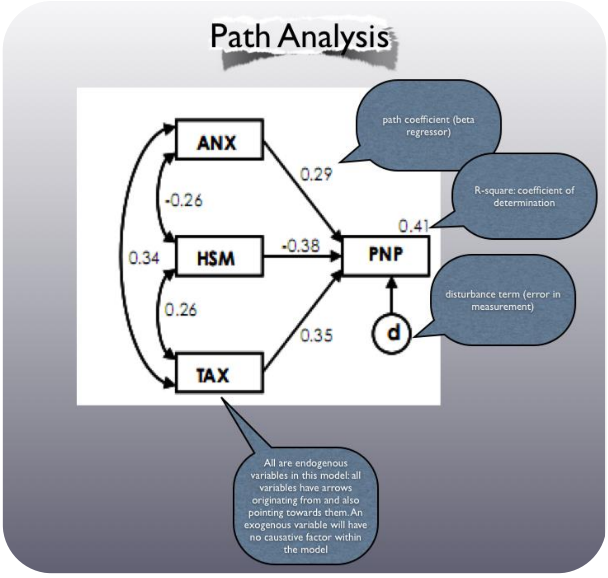
The **grasp reflex** is obtained when the examiner's hand is gently inserted into the palm of the patient's hand. A distraction such as ongoing conversation with the patient is useful. The palmar surface is stroked or simply touched. The flexor surfaces of the fingers may be stimulated also by the examiner's fingers. The stimulus should be in a distal direction. With a positive response, the patient grasps the examiner's hand with variable strength and continues to grasp as the examiner's hand is moved. Ability to release the grip voluntarily depends on the activity of the reflex; some patients can do so readily, while others can even be lifted off the bed, since the grasp has such power. A variant is to stroke the flexor surfaces of the patient's fingers. With positive responses, the fingers will curve much like a bird's claw and hook the examiner's fingers or hands. Some clinicians maintain that the reflex is brought out more easily if the patient is lying on the side with the hand to be tested uppermost.

A **foot grasp reflex** can be elicited by stroking gently the plantar surface medially with a blunt object such as the handle of a reflex hammer. The lateral surfaces of the foot bend as if to make a cup out of the plantar surface. The toes adduct; there is hollowing of the sole with some wrinkling of the skin. If the toes also flex, this is called the tonic foot response. In patients who also have the **Babinski** reflex, the Babinski can usually be elicited more laterally than the grasp.

126. In the path analysis, **is it a strong association when the beta value is high**?

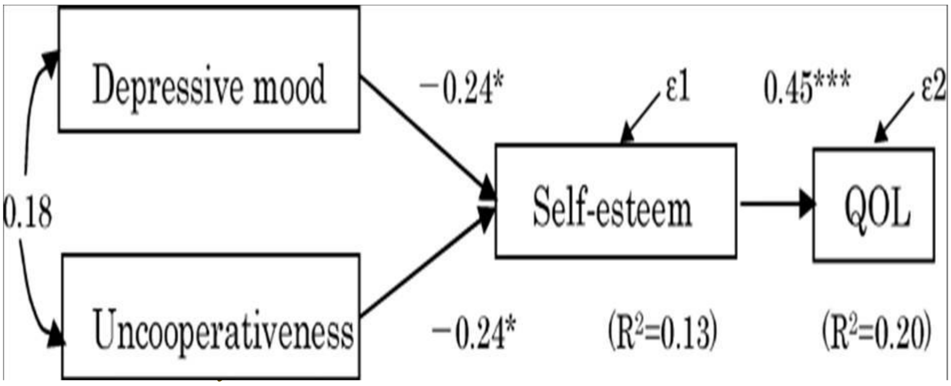
In statistical terms, a **high beta value indicates a strong association between the independent variable (predictor) and the dependent variable (outcome).** The beta value represents the change in the dependent variable for a one-unit change in the independent variable.

So, a **high beta value suggests a more substantial impact or influence of the independent variable on the dependent variable**. However, the interpretation of "high" depends on the context of your specific analysis and the range of values in your dataset. It's important to consider other factors such as statistical significance and the practical significance of the results.



Check Q. 31-35 in Fresh question 2019 SPMM

[SPMM]This figure represents which of the following statistical analyses:



B. Path analysis
C. Root cause analysis
D. Survival analysis
E. Economic analysis

Path analysis is an extension of multiple regression. It goes beyond regression in that it allows for the analysis of more complicated models. In particular, it can examine situations in which there are several final dependent variables and those in which there are chains" of influence, in that variable A influences variable B, which in turn affects variable C. Despite its previous name of "causal modelling," path analysis **cannot be used to establish causality** or even to determine whether a specific model is correct; it can only determine whether the data are consistent with the model".

127. Pain while wearing shoes was **allodynia**, right? Not hyperalgesia? the questions were the definitions Look Q. 19

128. Tics ongoing for 6 months, **transient** or non-specified?

129. Question about the link of rem sleep behavioural disorder and Lewy body dementia?

[SPMM]In which of the following dementias is REM sleep behaviour disorder a core feature?

Select one:

A. Lewy body dementia
B. Alzheimer's disease
C. Vascular Dementia
D. Fronto-temporal Dementia
E. Mixed Dementia

In rapid eye movement (REM) sleep behaviour disorder, 'dream-enacting behaviours' are reported. The affected individual physically acts out vivid, unpleasant dreams with vocal sounds and sudden, violent, arm and leg movements during the REM sleep.. It is a core feature of Dementia with Lewy bodies (DLB), and can precede cognitive decline by several years.

Core clinical features of dementia with Lewy bodies (DLB): **REM-PCV**

1. Fluctuating **C**ognition with pronounced variations in attention and alertness
2. Recurrent **V**isual hallucinations that are typically well formed and detailed
3. **REM** sleep behaviour disorder, which may precede cognitive decline
4. Spontaneous cardinal features of **P**arkinsonism (bradykinesia, rest tremor, and/or rigidity)

Supportive clinical features of DLB:

1. Severe sensitivity to antipsychotic agents
2. Postural instability Repeated falls
3. Syncope or other transient episodes of unresponsiveness
4. Severe autonomic dysfunction (e.g. constipation, orthostatic hypotension, urinary incontinence)
5. Hypersomnia
6. Hyposmia - decrease sense of smell
7. Systematized delusions
8. Hallucinations in other modalities
9. Apathy, anxiety, and depression

130. Question about a Lady with Mixed Anxiety depression on Sertraline & Quetiapine- having bodily movements, remembers episode...

A. Dissociative? I said dissociative. She was fully conscious.

B. Complex partial seizures were also in the options

- Quetiapine **moderate** risk of reducing seizure threshold and Sertraline **low** risk
- We agreed that If the question mentions any postictal symptoms could be complex partial otherwise might be dissociative ??

[SPMM] Mrs. Vaughn is a 38-year-old woman with a diagnosis of epilepsy. She has recently presented to her GP with symptoms of low mood, inability to enjoy her usual activities and poor sleep. Which of the following is most likely to increase the risk of developing depression among people with epilepsy?

Select one:

A. Family history of epilepsy
B. Generalised seizures
C. Early age at diagnosis
D. Complex partial seizures
E. Valproate use

Complex partial seizures and absence of secondary generalized tonic–clonic seizures are risk factors for depression. Treatment with clonazepam, frequent hospitalizations (drug resistance) and lack of occupational activity are additional significant contributing factors.

131. What medication lowers ECT seizure duration? **Benzo and anti-convulsant. See Table Q. 52**

132. Also another MCQ about types of sampling, they arranged them based on gender. It said they allocated randomly chosen subjects to groups based on their gender and diagnosis or smth like that, didn't it? **Minimisation**

Stratification: When Analysing results, tabulate data for various levels/categories of exposure to confounder separately and analyse the variable influence; e.g. analyse the association of antipsychotics and hip fractures in three different age groups individually. Thus, Subgroups are created for analysis. **Done during stage of analysis.**

133. Primitive defence mechanism. Introduction, **repression**, Identification, undoing, introjection **Look Q13**

[SPMM] Which of the following is a primary defence mechanism?

A. Displacement
B. Repression
C. Idealisation
D. Intellectualisation

134. Two arm of study must be equal, **block randomization**, minimization.

[SPMM]A researcher wants to make sure that the size of the control and two intervention groups are more or less equal so that statistical deductions can be meaningful. Which of the following methods of randomisation could ensure this?

Select one:

A. Block randomisation
B. Stratification
C. Cluster randomisation
D. Quasi-randomisation
E. Simple randomisation

Block randomization refers to a method where randomization occurs in a set of specified numbers, say blocks of 6 etc., to ensure equal number distribution between two arms. Larger the block size, lesser the ability to predict allocation.
In cluster randomization, the unit of randomization and analysis is various centres or catchment-zones and not individual patients.
Stratification is classifying to subgroups before randomising within each subgroup to ensure impartial allocation of subgroup factors.
Quasi-randomisation refers to 'randomizing' using even/odd numbers of the date of birth, day of the week patient was seen, etc.

135. Inheritance of down for both parents. **Robertsonian**, 21trisomy.

[SPMM] In Down syndrome, which of the following mechanisms have an equal chance of inheritance from either paternal or maternal genes?

Select one:

A. **Robertsonian translocation**
B. Mosaicism
C. Trisomy 21 – 47XX + 21
D. Trisomy 21 - 47XY + 21
E. Partial trisomy

Partial or complete trisomy 21 (that is, the presence of part of or a complete supernumerary chromosome 21) is the genomic cause of Down syndrome (DS). Most cases of DS are not inherited. However, people with translocation DS can inherit the condition from an unaffected parent **(either paternal or maternal).** Translocation accounts for trisomy 21 in ~5% of affected individuals, usually t(14;21) or t(21;21). A free trisomy 21 is present in 95% of individuals with DS and results from an error in maternal meiosis I (~66%) or meiosis II (~21%); paternal meiosis I (~3%) or meiosis II (~5%); or mitosis, after the formation of the zygote (5%). Advanced maternal age has been associated with chromosome 21 segregation errors in both maternal meiosis I and meiosis II. Mosaicism for trisomy 21 is not inherited and occurs in ~2% of individuals with DS. Partial trisomy 21 is rare and is associated with a range of symptoms that vary according to the length of the partial triplication of chromosome 21.

136. Hypothesis in psychotherapy, psychodynamic or **systemic**

Milan-Systemic therapists see the family as a self-regulating system which controls itself according to rules formed over a period of time through a process of trial and error. They are interested in the rule-maintaining characteristics of communication and behaviours, and assume that the way to eliminate a symptom is to change the rules. An interview consists almost entirely of questioning of the family by the therapist. Questioning is a recursive and circular process, with each question building upon the family's response to previous questions. Emphasis is placed on exploring differences between family member's behaviours, emotional responses and their beliefs at differing points in time. **Key terms include: hypothesising, neutrality, positive connotation, paradox and counter-paradox, circular and interventive questioning, and the use of reflecting teams.**

137. Which is a dynamic factor?

A. **Insight**
B. Previous offences
C. Gender
D. Age

Risk factors for any untoward incident (suicide, crime or violence) can be categorized as:

- **Static risk factors**: These are fixed and historical: e.g. family history of suicide. They cannot be modified.
- **Stable risk factors**: These are long term, enduring issues but are modifiable to some extent and not fixed: e.g. diagnosis of personality disorder.
- **Dynamic risk factors** :fluctuate markedly in both duration and intensity: the e.g. presence of acute anxiety symptoms or akathisia. A dynamic risk factor can act synergistically and multiply the effect of underlying static and stable risk factors if not addressed promptly.

  Certain dynamic factors may occur only in future (e.g. upon discharge, a patient may feel helpless).

[SPMM]Mr Daniel Alwin is a 28-year-old man who has been diagnosed with dissocial personality disorder. He has recently been released from prison after serving a sentence for actual bodily harm. He was reviewed by a psychologist prior to release to determine his risk to others in the future. Which of the following is most accurate regarding short to medium term risk assessment?
Select one:

A. Risk factors for violent and sexual offending are the same
B. Static risk factors are most important to consider
C. The relevant risk factors for male and female offenders are the same
D. Substance misuse history is unlikely to modify risk
E. Dynamic risk factors are most important to consider

Dynamic risk factors, such as substance misuse, engagement with relevant interventions and expression of anger, are thought to be more relevant for risk assessment in the short term. Static risks, such as childhood experiences, previous violent incidents and demographic factors, are thought to be more relevant over the longer term.

138. There was a question about where the researcher does the survey himself or something and it asked about the bias: selection bias or **information bias** or something? Same person does the screening and diagnostic interview, so when they do the diagnostic interview they are already biased due to knowing what they scored in the screening : **information bias**

Interviewer/Observer bias: Interviewer or observer knowledge about in-question hypothesis and disease or/and exposure can take effect on collection and registry of data.

Information bias: (when gathered information about exposure, outcome or both is not correct and there was an error in measurement)

A study relies on a single psychiatrist to establish the diagnosis of all the participants. Which of the following types of bias would potentially result from this methodology?

A. Confounding bias
B. Information bias
C. Berkson's bias (hospital admission bias)
D. Neyman bias (incidence prevalence bias)
E. Healthy worker effect

139. Was the prison suicide risk question discussed?
Remand, sentenced, has kids. Substance use was also option

[SPMM]Which of the following is true of suicide in prison?

Select one:

A. Being convicted of homicide increases risk

B. Being in a single cell reduces risk
C. Men and women are at equal risk
D. Being sentenced increases risk when compared with remand status
E. Physical and mental co-morbidities do not increase risk

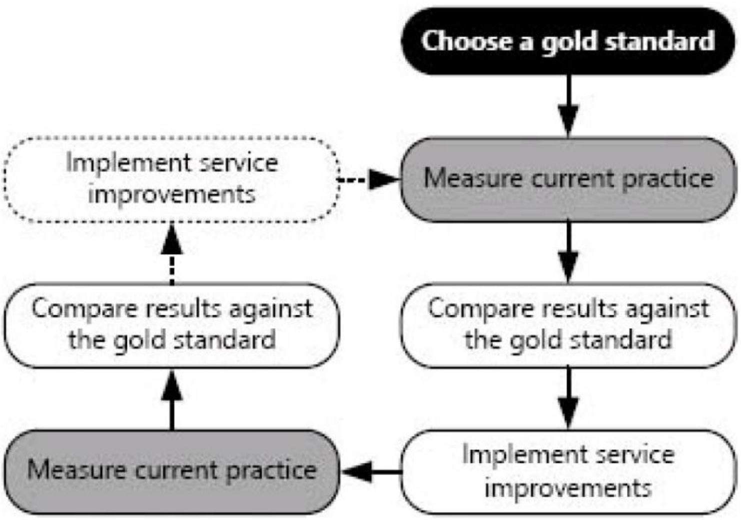
Deaths by suicide among people in prison have long been shown to occur at higher rates than among general populations of similar ages. Suicide rates in **male prisoners are 3-8 times** higher than in the general population, whereas the rate in female prisoners is typically **more than 10 times higher**. **The strongest clinical factors associated with suicide are suicidal ideation during the current period in prison, a history of attempted suicide, and current psychiatric diagnosis. Institutional factors associated with suicide include occupation of a single cell and having no social visits**. **Being sentenced is associated with a reduced suicide risk when compared with detainee or remand status.** Other criminological factors associated with increased risk include serving a life sentence and being convicted of a violent offence, in particular homicide.

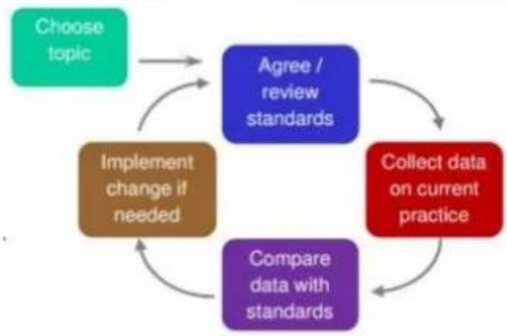
Remand prisoners, young offenders, and those with a history of substance misuse and violent offences are also at higher risk of suicide and occupied single cells.

Q. Which of the following factors has been shown to increase the risk of suicide in prisoners?
A. Having children
B. Being female
C. Being older
D. Being unmarried
E. Being on remand

Being married in prison is a risk factor while it is protective factor outside prison

140. There was a question on Audit procedure- **Change not implemented**





[SPMM]A consultant asks you to complete an audit on a prescribing nicotine replacement therapy for new inpatients. You identify the Trust's Drug & Therapeutics Guideline as the gold standard and collect the data. You then present it to your consultant colleagues. Six months later you reaudit the process to determine if there has been an improvement. What important aspect of the audit process has been omitted?
Select one:

A. Publishing the results in a peer-reviewed journal
B. Presenting the findings at a national level
C. Using NICE guidelines
D. Developing a hypothesis
E. Initiating change after the first cycle

Before re-auditing, one must **implement changes** to see if they have made a difference.

[SPMM]Audit is a central part of modern medical practice. Which of the following is the main purpose of an audit?
Select one:

A. To check a service is performing at a given standard
B. Clinical governance
C. To determine if an intervention works
D. To improve services
E. To provide a basis for research

Clinical audit is a process that has been defined as "a quality improvement process that seeks to improve patient care and outcomes through systematic review of care against explicit criteria and the implementation of change". It involves the comparison of existing practice against an appropriate standard.

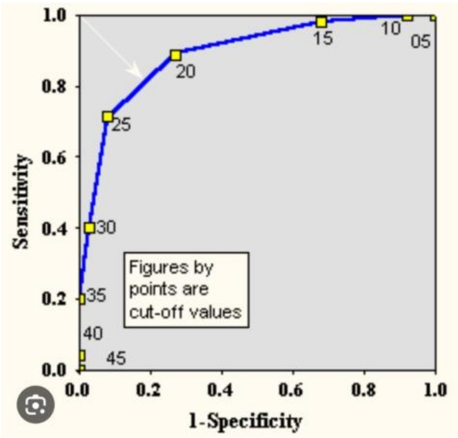
141. Question about girl in over dose with respiratory problem, but I don't remember options,

A. **Naloxone**,
B. Buprenorphine,
C. Antidote:

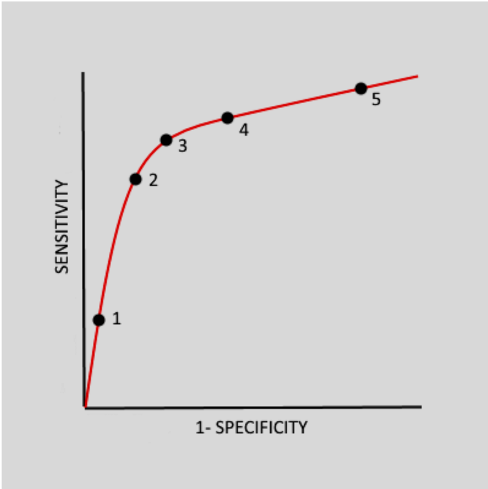
Was this the one specifying ' use of accessory muscles', or was it a different question? **Naloxone**
We think it is a case of opioid toxicity, however question is not clear.

Methadone (opioid agonist) or buprenorphine (partial agonist at the μ opioid receptor) should be offered as the first-line treatment in opioid detoxification (there is no evidence favoring one over the other - Maudsley 13th Ed)

142. Roc curve questions
1) asking which point is the best cut-off for a diagnostic test for delirium unsure between 3 and 4



[Psych-Mentor] A frustrated psychiatrist creates a new screening tool to aid in the diagnosis of schizophrenia. The tool consist of just five questions and can be done in 10 minutes. He does a study on the tool versus the gold standard method of diagnosis to see how good the screening tool performs. He sets a number of cut-offs for the tool to help decide how many positive question responses are needed prior to a diagnosis of schizophrenia being made. Following this he constructs a ROC curve to help him see how the cut-off levels compare. Based on the ROC curve, which cut-off point provides the best compromise between sensitivity and specificity?



A. Cut-off point 1
B. Cut-off point 2
C. Cut-off point 3
D. Cut-off point 4
E. Cut-off point 5

The dot closest to the top left hand corner is the one with the best trade off between sensitivity and specificity.

[SPMM] Ability of Predementia AD Scale (PAS) to predict dementia in mild cognitive impaired subjects was assessed. Two samples with baseline PAS scores were followed up either for 2 years or for 5 years. The present study investigated whether stepwise scoring of the PAS can reduce the number of subjects who need to undergo elaborate or expensive diagnostic procedures for predicting outcome. (Visser et al. Journal of Neurology 2002, 249: 312-319). Use the attached graph to answer the following question.

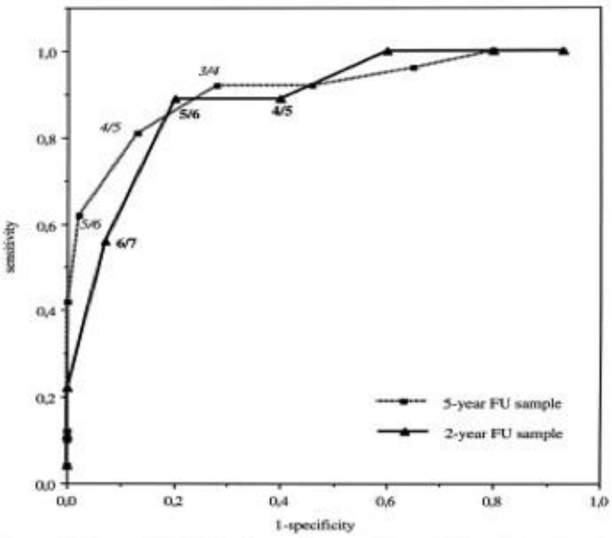


Fig. 1 ROC curve of PAS.The italic cut-off scores refer to the 5-year FU sample and the bold cut-off scores to the 2-year FU sample.

The cut-off value with **highest** sensitivity with respect to the 5 years follow-up sample is:

A. 3/5
B. 5/6
C. 3/4
D. 6/7
E. 4/5

The cut-off value with **lowest** sensitivity with respect to 2 years follow-up sample is:

A. 3/5
B. 5/6
C. 3/4
D. 6/7
E. 4/5

The value which gives least number of **false positives in 2 years sample is**

A. 3/5
B. 5/6
C. 3/4
D. 6/7
E. 4/5

To answer this question consider the following Specificity = true negative / total not diseased (TNR = true negative rate is same as specificity). The expression (1 - specificity) = 1 - (true negative / total not-diseased); As true negatives + false positives together constitute all 'non diseased', 1 - (true negative / total not-diseased) can be rewritten as = false positives / total non-diseased = false positive rate (FPR). In other words, in an ROC curve, the X axis can be considered as False positive rate, the least number of which will be given by 6/7 for the 5 years curve.

- **Sensitivity = true positive / total diseased**
- **Specificity = true negative / total non diseased**

143. That question on reoffending was what reduces it, **Appearance in court** was also an option.

[SPMM] Mr Northgate is a 48-year-old man who has been convicted for exhibitionism after exposing himself in a local park. His wife is supporting him and wishes to know what would stop him from acting in this way again. Which of the following factors is associated with him not reoffending?

Select one:

A. Late onset of offending
B. Alcohol misuse
C. Having to appear in court
D. Having exposed himself to people of different ages
E. Previous convictions for theft

Earlier onset of offending is associated with recidivism. Many exhibitionists have prior criminal histories and around 30% will go on to commit other sexual offences. Victimizing individuals of various ages is associated with an increased risk of reoffending. Alcohol misuse and previous convictions for other offences are associated with recidivism.

144. Were there 3 questions on Qualitative research?

Qualitative Research 1
A qualitative research study is conducted to obtain the views of patients with schizophrenia regarding the nature of mental health care they received. From the list of options choose the most appropriate methods to be used in this study from the descriptions below:

You ask psychologists and social workers who provide care to the participating patients to be the study subjects to achieve a wider perspective

You include an independent rater to look at the data collected when analysing it.

You want a systematic approach of analysing the data alongside the collection.

Method used to reach adequate sample size

Choose...
focus group
ethnography
Grounded theory
snowball sampling
Structured interviews
Semi-structured interviews
respondent validation
Triangulation
Theoretical saturation
Purposive sampling

Your answer is incorrect.

Explanation:

To reach adequate sample size and stop collecting data, the point of theoretical saturation is employed in qualitative research. The data is iteratively analysed and at the point where no more new categories emerge, the data collection ceases. This point is called as theoretical saturation.

Triangulation compares the results from either two or more different methods of data collection or two or more data sources. The researcher looks for patterns of convergence to develop or corroborate an overall interpretation. In investigator triangulation, more than one investigator is involved, and findings are cross-validated.

Purposive sampling is one of the most common sampling strategies in qualitative research. (Note that snowball sampling and quota sampling are also used often). In purposive sampling the participants are selected according to a preselected criteria relevant to a particular research question - hence it is not random. Sample sizes, which may or may not be fixed prior to data collection, depend on the resources and time available, as well as the study's objectives. Purposive sample sizes are often determined on the basis of theoretical saturation (the point in data collection when new data no longer bring additional insights to the research questions). Purposive sampling is, therefore, most successful when data review and analysis are done in conjunction with data collection.

Grounded theory refers to constant comparison of the collected data alongside the data collection (iterative analysis)

The correct answer is: You ask psychologists and social workers who provide care to the participating patients to be the study subjects to achieve a wider perspective → Purposive sampling, You include an independent rater to look at the data collected when analysing it. → Triangulation, You want a systematic approach of analysing the data alongside the collection. → Grounded theory, Method used to reach adequate sample size → Theoretical saturation

Sampling techniques for qualitative research
A researcher wants to explore service users' perceptions and attitudes associated with the prescription and administration of clozapine in community mental health settings. He chooses to use one of the following sampling methods. Choose the correct term describing each approach from the list below.

The researcher chooses to offer a semi-structured interview to the patients who attend clozapine monitoring clinic once every three months. All active community clozapine users are registered to attend this clinic.

The researcher pursues a set of service users who experienced an unusual increase in sweating with clozapine. Another service user who shared the same effect introduced these patients to the researcher.

The researcher specifically attempts to include some highly satisfied service users, some poorly compliant users, and some users who discontinued early in the course and some users who continue to have psychotic symptoms despite high degree of concordance.

The researcher recruits four service-user experts who have wide experience of talking to clozapine users in the locality.

Choose...
Snowballing
Random sampling
Systematic sampling
Multistage sampling
Concordant sampling
Purposive sampling
Convenience sampling
Intensity sampling

Convenience (or opportunistic) sampling is a technique that uses an open period of recruitment that continues until a set number of subjects, events, or institutions are enrolled. Here, the selection is based on a first-come, first-served basis. This approach is used in studies drawing on predefined populations such as participants in support groups or medical clinics.

Second, purposive sampling is a practice where subjects are intentionally selected to represent some explicit predefined traits or conditions. This is analogous to stratified samples in probability-based approaches. The goal here is to provide for relatively equal numbers of different elements or people to enable exploration and description of the conditions and meanings occurring within each of the study conditions. The objective, however, is not to determine the prevalence, incidence, or causes.

Third, snowballing or word-of-mouth techniques make use of participants as referral sources.

Participants recommend others they know who may be eligible.
Fourth, case study samples select a single individual, institution, or event as the total sample. A variant is the key-informant approach or **intensity sampling** (Patton 1990) where a subject who is an expert in the topic of study serves to provide expert information on the specialized topic. When qualitative perspectives are sought as part of clinical or survey studies, the purposive or case study sampling techniques are generally the most useful

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145. Also, was there a question on reliability in the stats section?

Q. Which one among the following is a measure of internal consistency of a test?

Select one:

A. Content validity
B. Test-retest reliability
C. Inter-rater reliability
D. Face validity
E. Split half reliability

Split-half reliability is a measure of internal consistency reliability. The items on the scale are divided into 2 halves and the resulting half scores are correlated in reliability analysis. High correlations between the halves indicate consistency in reliability analysis.

Validity and Reliability
Match the following scenarios to their description of validity or reliability:

1. Multisource feedback in for doctors in Foundation Year 2 (F2) was in line with multisource feedback in Foundation Year 1 (F1)

2. Two respondents on multisource feedback gave similar positive reports on the same F1 trainee.

3. Two radiologists reported the same degree of hippocampal atrophy and this was associated with a diagnosis of Alzheimer's dementia

4. A new screening questionnaire for depression was being evaluated and compared to the gold standard that was administered at the same time to see if it could still diagnose depression with fewer items or questions.

Inter-rater reliability
Test re-test reliability
Concurrent validity
Construct validity
Content validity
Predictive validity

Explanation: Inter-rater reliability is used to assess the degree to which different raters/observers give consistent estimates of the same phenomenon. Having good test re-test reliability signifies the internal validity of a test and ensures that the measurements obtained in one sitting are both representative and stable over time. Concurrent validity is a measure of how well a particular test correlates with a previously validated measure. Predictive validity involves testing a group of subjects for a certain construct, and then comparing them with results obtained at some point in the future. Construct validity refers to whether a scale or test measures the construct adequately. It is a broader concept and is best opted for if other options are not specific enough.

1. Multisource feedback in for doctors in Foundation Year 2 (F2) was in line with multisource feedback in Foundation Year 1 (F1) – **Predictive validity**

2. Two respondents on multisource feedback gave similar positive reports on the same F1 trainee. - **Inter-rater reliability**

3. Two radiologists reported the same degree of hippocampal atrophy and this was associated with a diagnosis of Alzheimer's dementia - **Inter-rater reliability**

4. A new screening questionnaire for depression was being evaluated and compared to the gold standard that was administered at the same time to see if it could still diagnose depression with fewer items or questions. - **Concurrent validity**

Q. A psychiatrist wants to examine the validity of MRCPsych theory exams that purportedly test the knowledge base of a psychiatric trainee. She chooses to examine the degree of association between paper A scores and paper B scores. Which of the following is she testing?

Select one:

A. Content validity
B. Divergent validity
C. Face validity

D. Convergent validity
E. Divergent validity

Convergent validity is a part of construct validity. **Two measures of the same construct** should be, in theory, related to each other. We can test this correspondence or convergence using correlation metrics (Note that the construct examined by the two measures is the same - theoretical knowledge base). **Predictive validity** refers to the ability of a test to predict future group differences, according to current group differences in score. E.g., can paper A score predict ARCP outcome or CASC scores (note that the two constructs examined here are not the same, they may be related).

Q. A primary care depression screening tool is evaluated by testing whether those who are diagnosed as 'depressed' using this scale are also found to be 'depressed' using Beck's depression inventory, Hamilton's depression scale and clinician assessment. Which of the following is measured here?
Select one:

A. Internal consistency
B. Divergent validity
C. Convergent validity
D. Predictive validity
E. Inter-rater reliability

Convergent and divergent validity together constitute construct validity

- Two measures for the same thing – **Convergent Validity (checking hands**)
- Two measures for 2 different things – **Divergent** – **dis**agreement
- Face validity – Like snap shot of something (looking for a word in a scale for example)

146. Purpose of UDS during treatment of substance abuse

Chat GPT In the treatment of substance abuse, urine drug screens serve several important purposes. They help healthcare professionals **monitor a patient's progress**, ensure compliance with treatment plans, and detect any potential relapses. Regular urine drug testing provides objective evidence of a patient's substance use, helping to tailor and adjust their treatment strategies accordingly. It also fosters accountability, facilitating open and honest communication between healthcare providers and individuals undergoing substance abuse treatment.

147. CI for clozapine initiation in the community.

Guidelines for the initiation of clozapine for patients based in the community

Contra-indications to community initiation

- History of seizures, significant cardiac disease, unstable diabetes, paralytic ileus, blood dyscrasia, neuroleptic malignant syndrome or other disorder that increases the risk of serious side-effects (initiation with close monitoring in hospital may still be possible)
- Previous severe side effects on titration of clozapine or other antipsychotics
- Unreliable or chaotic life-style that may affect adherence to the medication or the monitoring regimen
- Significant abuse of alcohol or other drugs likely to increase the risk of side-effects (e.g. cocaine)

Suitability for community initiation (answers should all be yes)

- Is the patient likely to be adherent with oral medication and to monitoring requirements?
- Has the patient understood the need for regular physical monitoring and blood tests?
- Has the patient understood the possible side-effects and what to do about them (particularly the rare but serious ones)?
- Is the patient readily contactable (e.g. in the event of a result that needs follow-up)?
- Is it possible for the patient to be seen every day during the early titration phase?
- Is the patient able to collect medication every week or can medication be delivered to their home?
- Is the patient likely to be able to seek help out-of-hours if they experience potentially serious side-effects (e.g. indicators of myocarditis or infection such as fever, malaise, chest pain)?

The BNF advices caution in the following circumstances:

- Prostatic hypertrophy
- Susceptibility to angle-closure glaucoma
- Adult over 60 years

Contraindications Go to:

Clozapine is contraindicated in patients with serious hypersensitivity reactions to clozapine or any component of the formulation.

FDA states the following boxed warnings:

- Neutropenia (due to the risk of agranulocytosis)
- Orthostatic hypotension, bradycardia, and syncope
- Seizures
- Myocarditis and mitral valve incompetence
- Increased mortality in dementia-related psychosis in elderly patients (risk of a cardiovascular event)[29]

148. Something to do with cocaine and intoxication? Don't remember exactly
Ischemic changes – tachycardia and tachypnoea

Drug	UK Street names	Effects	Withdrawal features
Opioids	Heroin = Smack, horse, junk, skag, brown, Harry, H	Euphoria, drowsiness, constipation, pupillary constriction, respiratory depression	Piloerection, insomnia, restlessness, dilated pupils, yawning, sweating, abdominal cramps
Amphetamine and cocaine	**Amphetamines = Speed, meth, ice, uppers, whizz. Cocaine = Snow, crack, coke, rock**	**Increased energy, insomnia, hyperactivity, euphoria, paranoia, reduced appetite**	**Hypersomnia, hyperphagia, depression, irritability, agitation, vivid dreams, increased appetite**
MDMA (ecstasy)	MDMA = XTC, E, Eccies, pills, Adam, Mandy	Increased energy, increased sweating, jaw clenching, euphoria, enhanced sociability, increased response to touch	Depression, insomnia, depersonalisation, derealisation
Cannabis	Cannabis = Marijuana, weed, pot, grass, joint, mull, cone, ganja, hash, chronic	Relaxation, intensified sensory experience, paranoia, anxiety, injected conjunctivae	Insomnia, reduced appetite, irritability
Hallucinogens	LSD = Trips, Tripper, Tab, Stars, Smilies, Rainbows, Paper	Perceptual changes, pupillary dilation, tachycardia, sweating, palpitations, tremors, inco-ordination	No recognised withdrawal syndrome
Ketamine	Ketamine = Vitamin K, Super K, Special K, K, donkey dust	Euphoria, dissociation, ataxia, hallucinations, muscle rigidity	No recognised withdrawal syndrome

149. Torsade de pointes causing drug:

Class	Examples
Antiarrhythmics	Disopyramide, procainamide, quinidine, sotalol
Macrolides	Azithromycin, clarithromycin, erythromycin
Fluoroquinolones	Ciprofloxacin, levofloxacin, moxifloxacin
Antifungals	Fluconazole, ketoconazole, pentamidine, voriconazole
Antipsychotics	Haloperidol, thioridazine, ziprasidone
Antidepressants	Citalopram, escitalopram,
Antiemetics	Dolasetron, droperidol, granisetron, ondansetron
Opioids	Methadone
Miscellaneous	Cocaine, cilostazol, donepezil

150. Some question related to? Narcolepsy

Q. Which of the following is true regarding narcolepsy?

A. Onset is usually in the fourth decade
B. Following naps patients with narcolepsy do not feel refreshed
C. Antidepressants are used to help with cataplexy
D. Sleep paralysis usually occurs in non REM sleep
E. Cataplexy is characterised by generalised increased muscle tone

Narcolepsy usually starts at 15-30 years of age

Q. Modafinil is indicated in which of the following conditions?

A. Obsessive compulsive disorder
B. Panic disorder
C. Narcolepsy
D. Schizophrenia
E. Dementia

Q. Which of the following is associated with narcolepsy?

A. Prolonged REM latency
B. Unrefreshing sleep attacks
C. 2% prevalence
D. Waxy flexibility
E. HLA-DR2

Narcolepsy is associated with cataplexy (NOT catalepsy / waxy flexibility – the latter are catatonic features). It is a disabling sleep disorder affecting 0·02% of adults worldwide. It is characterised by severe, irresistible daytime sleepiness and sudden loss of muscle tone (cataplexy), and can be associated with sleep-onset or sleep-offset paralysis and hallucinations, frequent movement and awakening during sleep, and weight gain. Sleep monitoring during night and day shows rapid sleep onset and abnormal, shortened rapid-eye-movement sleep latencies. The onset of narcolepsy with cataplexy is usually during teenage and young adulthood and persists throughout the lifetime.

Pathophysiological studies have shown that the disease is caused by the early loss of neurons in the hypothalamus that produce hypocretin, a wakefulness-associated neurotransmitter present in cerebrospinal fluid. The cause of neural loss could be autoimmune since most patients (>90%) have the HLA DQB1*0602 allele that predisposes individuals to the disorder. **The frequency of HLA-DR2 is also increased in narcolepsy** though not as high as HLA DQB1*0602. Treatment is with stimulant drugs to suppress daytime sleepiness, antidepressants for cataplexy, and ? hydroxybutyrate for both symptoms

151. Aggressive kid not responding to non-pharmacological measures. **Risperidone**

[SPMM]A 9-year-old girl with ADHD and frequent aggressive outbursts has not responded to trials of methylphenidate or guanfacine. A trial of which of the following medications is now indicated?

Select one:

A. Pemoline
B. Clonazepam
C. Olanzapine
D. Imipramine
E. **Risperidone**

The initial treatment of children and youth with ADHD with disruptive and aggressive behaviour includes the use of psychosocial interventions and psychostimulants to treat core ADHD symptoms. Psychostimulants have demonstrated efficacy for overt and covert aggression, as well as oppositional behaviour and conduct problems in children with ADHD. If these strategies are ineffective, there is evidence to support the use of **atomoxetine, guanfacine and clonidine (CAG)** for oppositional behaviour and conduct problems associated with ADHD. **As the side effect burden of these medications is lower than with antipsychotics, their use is preferable over risperidone**. If adequate symptom control is not achieved with the use of atomoxetine, guanfacine or clonidine, or they are contraindicated or intolerable, a trial of risperidone may be attempted. If risperidone is ineffective there is very little evidence to support the use of other antipsychotics**. In general, the use of second-generation antipsychotics for the treatment of ADHD is not recommended**

152. History of Seizures? Citalopram?
Sleep paralysis Risperidone
What antidepressant to use with seizures?

Table 10.5 Psychotropics in epilepsy — Maudsley Guidelines P. 789

Safety in epilepsy	Drug	Comments
Antidepressants		
Low risk – good choices	SSRIs	Recommended in PWE.[14,18] SSRIs may be anticonvulsant at therapeutic doses[13] but pro-convulsant in overdose.[31] SSRIs with the lowest risk of interactions with antiseizure medications are generally preferred (citalopram/escitalopram, followed by sertraline).[14,18,32,33] Escitalopram is preferred over citalopram in PWE (lower risk of seizures in overdose).[34] Others have low risk of seizures (e.g. fluoxetine[34]) but drug interactions with antiseizure medications should be considered.[14,18] Fluoxetine may be less likely to provoke seizures in older people than escitalopram or citalopram.[35] Some evidence that sertraline is safe and effective in PWE[36]
	Mirtazapine	Recommended in PWE.[18,37] Not known to be proconvulsive[24]
	Duloxetine	Recommended for PWE.[11,18] Risk of seizures is probably negligible[34,35]
Probably low risk – use with caution (limited evidence)	Agomelatine	Not known to be proconvulsive.[38] Anticonvulsant in animal models[34]
	MAOIs	Not known to be pro-convulsive at therapeutic doses.[34] Low risk of seizures in overdose[17]
	Moclobemide	Not known to be proconvulsive.[34] Anticonvulsant in animal models[34]
	Reboxetine	Small open label study suggests no problems in PWE[39]
	Vortioxetine	Not known to be proconvulsive[34,40] but no experience in PWE[34]
Moderate risk – care required	Lithium	Low risk of seizures.[34] Anticonvulsant in animal models.[34] However, limited data showing increases or decreases in seizures frequency in PWE.[34] For bipolar, consider anticonvulsant mood stabilisers[41]
	Trazodone	Limited data suggest some risk of seizures[34,42]
	Venlafaxine	Effective in PWE[11] and has been recommended[18] but mixed evidence on seizure risk[34]
	Vilazodone	Limited data. Seizure exacerbation in a patient with epilepsy has been reported[34]
Higher risk – avoid (pro-convulsive at therapeutic doses[13])	Amoxapine	Several reports of seizures at therapeutic doses[42]
	Bupropion	Dose-related risk of seizures (particularly with instant-release formulations).[34] Risk is less with slow-release formulations at doses under 300mg/day[34]
	Maprotiline	Several reports of seizures at therapeutic doses[42]
	TCAs	Most TCAs are epileptogenic at higher doses (particularly clomipramine and amitriptyline[10,24,42]). Doxepin possibly lower risk (one small study in PWE).[34] SNRIs are preferred over TCAs in PWE[17]

(Continued)

153. Purpose of uds, is it to find out what drug they're using, to check what antidote to give??
: **Treatment compliance was one option, depends on the question.**

154. One where what is impaired in cocaine dependence? Attention?

[SPMM] 40-year-old man is a long term cocaine user. He is likely to have abnormalities in which of the following cognitive domains?

Select one:

A. Sustained Attention
B. Alertness and arousal
C. Abstract Reasoning
D. Executive Function
E. Verbal Fluency

Being a stimulant, cocaine enhances cognitive functions, especially response inhibition and psychomotor processing speed, acutely after administration, though long-term use impairs cognition. 30% of dependent users and even 12% of recreational users exhibit clinically relevant cognitive impairment (see Vonmoos et al., 2014). **The most impaired domains were sustained attention, response inhibition (impulsivity), verbal learning/memory, and working memory**. Abstinence helps to attain a level of cognitive performance level similar to non-users. Such recovery of cognition is not complete, as working memory impairments may continue despite 1-year of abstinence, especially in those who start cocaine use at an early age (Vonmoos et al., 2014).

[SPMM] Withdrawal: Effects of cocaine are short-lived - rapid cessation occurs due to rapid metabolism resulting in withdrawal effects that can lead to repeated use. A characteristic feature is **intense cravings** for cocaine with a striking lack of much physical withdrawal symptoms. **Dysphoria** (**crash**), **anhedonia**, **anxiety, irritability hypersomnolence**, and **sometimes agitation** are seen upon withdrawal. These effects usually end within **18 hours** though in heavy, dependent users this **can last up to a week, peaking in 3 days.**

Physical adverse effects of cocaine includes:

- Nasal perforation on snorting.
- Non-haemorrhagic cerebral infarctions.
- Subarachnoid, intraparenchymal, and intraventricular hemorrhages.
- TIAs
- Seizures (3 to 8% of A&E visits)
- Myocardial infarctions and arrhythmias

155. There was one question about prolonged grief and depression.

A. Feeling of anger
B. Sadness of mood
C. Lack of interest
D. Guilt
E. Suicidal ideation

[SPMM] Mrs Galloway's husband died of lung cancer 12 months ago. He had been severely unwell for two months prior to his death. She has come to see you and describes her struggles to you. Which of the following features is most suggestive of depression?

Select one:

A. Visual hallucinations of her husband
B. Excessive guilt feelings
C. Preoccupation with the deceased
D. Denial

E. Anger and irritability

Explanation: An excessive guilt feeling is most suggestive of depression. Feelings of guilt can be normal, after the death of someone close. But if they are strong and prolonged, then it is suggestive of depressive reaction. Visual hallucinations may also be part of normal grief, but if they are frequent, persistent, and disruptive to functioning then a psychotic illness should be considered.